Decory Edwards*

2026-02-25

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2 Literature Review

2.1 Trust

There are three strands of the trust literature of interest to this paper. First, this paper draws from the literature regarding the detereminants of trust. An example of this is Alesina and Ferrara 2000. Key takeaway is that the standard demographic control variables (age, race, education, gender) as well as living in a community with much heterogeneity in terms of either race or income.

This paper also seeks to contribute to the trust literature regarding the empirical relationship between measures of trust and economic performance. A notable example relevant for this paper is Guiso, Sapienza, and Zingales 2008 which finds that trust can lead to greater stock market participation.

A related work Guiso, Sapienza, and Zingales 2004 finds that measures of social capital and trust leads to greater use of financial instruments like checks, as well as poortfolio diversification away from holding cash and towards holding stocks and other assets.

A paper which serves as a key motivation for this work is Butler, Giuliano, and Guiso 2016. They find that the relationship between trust and income is humped shaped; an intermediate level of trust is associated with the maximal level of income.

Causal effects in the trust literature also inform this paper. Alsan and Wanamaker 2018 find that the 1972 public revelation of the Tuskegee Syphilis Study lead to lower trust in black males in medical institutions, ultimately leading to worse health outcomes over a lifetime.

Algan and Cahuc 2010 find that aggregate trust has a large positive causal effect on GDP per capita by showing that a measure of inherited trust by second generation immigrants in the U.S. is strongly correlated with aggregate trust in the origin country (and is persistent across generations).

2.2 Returns

An important work in the literature on empirical estimates of the rate of return is Fagereng et al. 2020 which characterizes the persistent component of returns for indivdiuals in Nowegian population data. This work also provides estimates of the distribution of returns within narrowly defined asset class, as well as for net wealth, which can serve as a useful benchmark for comparison.

Daminato and Pistaferri 2024 also documents the persistent component of returns in the PSID, and then use the data to calibrate an return-earnings process within a life cycle model of consumption-saving behavior.

3 Descriptive statistics

For the statistical analysis in this paper, I use the RAND Longitudinal version of the Household Retirement Survey (HRS) public data. I focus on the years 2002-2022; as this time horizon was filled with many economically relevant events (GFC, Covid), we can be creative regarding how to assess the accuracy of the data measured here. This is useful because, although there are many sources of income data to compare resuults to, there are less counterparts like this for return measures. The variables of interest fall into the categories: i) income, ii) wealth and portfolio composition, iii) returns, iv) trust, v) demographics, and vi) other controls.

3.1 Income

First, I use two measures of income: labor and total. Labor income is a narrow measure only capturing earnings and unemployment income. Total income is a more broad measure including retirement and capital income. First, it is clear that mean incomes are relatively flat over the period.

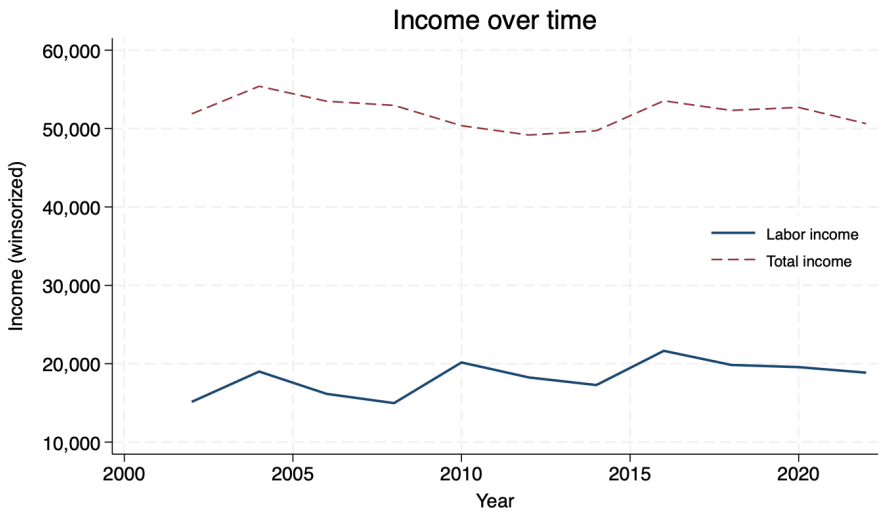

Figure 1: Income over time

Table 1: Mean income by year (real, winsorized)

Year	Labor income (mean $)	Total income (mean $)	Obs
2002	15,136	51,880	18,165
2004	19,003	55,399	20,129
2006	16,159	53,491	18,469
2008	14,976	52,959	17,217
2010	20,165	50,371	22,034
2012	18,250	49,179	20,554
2014	17,278	49,728	18,747
2016	21,644	53,540	20,912
2018	19,848	52,321	17,146
2020	19,568	52,703	15,723
2022	18,860	50,625	15,856

Real USD; winsorized at 1st and 99th percentile.

In particular, I pooled the observations of the survey together to get a sense of the distribution of incomes measured in the survey. This is captured by Table 3.1.

Table 2: Income (real, winsorized): summary statistics

Variable	Obs	Mean	SD	P50	P95	Min	Max
Labor income (real, winsorized)	204,952	18,330	37,032	0	97,924	0	215,261
Total income (real, winsorized)	204,952	52,004	68,036	29,103	177,498	0	457,548

Real USD, winsorized; summary over person-years.

An important finding in the literature on earnings in the U.S. is that it is generally hump-shaped over the lifecycle. It is a good sign then that Table 3.1 shows

Table 3: Mean income by age group (2022)

Age (midpoint)	Labor income (mean $)	Total income (mean $)	Obs
25	0	11,355	2
30	32,109	88,034	8
35	19,995	60,832	28
40	25,892	47,446	95
45	36,852	63,806	233
50	43,923	69,039	1,286
55	38,022	62,014	2,285
60	28,186	55,429	2,924
65	14,749	46,994	2,641
70	6,732	43,629	2,204
75	3,487	43,556	1,401
80	1,464	39,318	1,421
85	528	37,903	871
90	519	39,476	361
95	0	46,538	79
100	0	14,517	16
105	0	11,112	1
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Five-year age bins (e.g., 50 = 50–54). Real USD, winsorized.

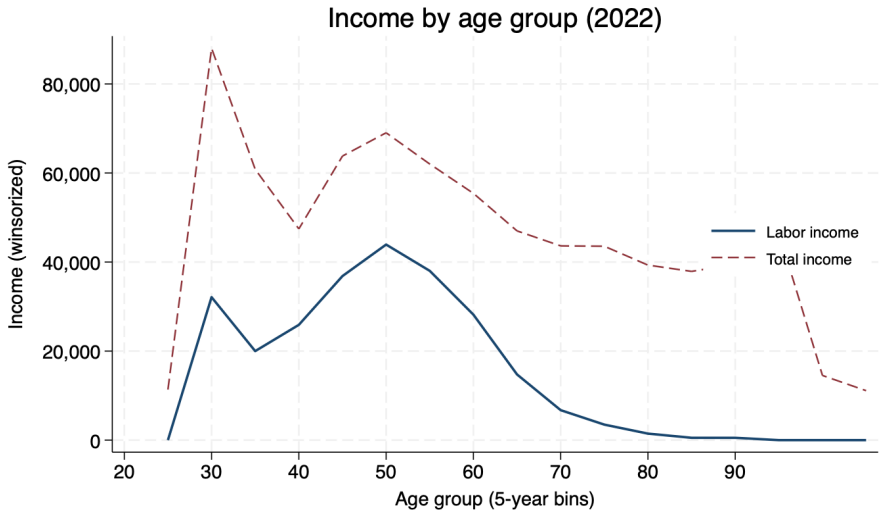

Figure 2: Income by age group (2022)

Another empirical finding in the literature on earnings is that on average individuals with more education earn higher income. The trend captured in Table 3.1 suggests that the earnings data is in line with one would respect regarding earnings for a representative survey in the U.S.[1]

Table 4: Mean income by education group (real, winsorized)

Education	Labor income (mean $)	Total income (mean $)	Obs
no hs	6,075	23,987	44,654
hs	12,590	39,601	63,826
some college	20,298	53,483	47,390
4yr degree	32,779	82,578	24,820
grad	37,445	104,066	23,091

Real USD, winsorized. no hs = <12y; hs = 12y; some college = 13–15y; 4yr = 16y; grad = 17+y.

[1] Although the HRS oversamples older households, I use the provided the respondent-level weights for this interpretation of the summary statistics of the data.

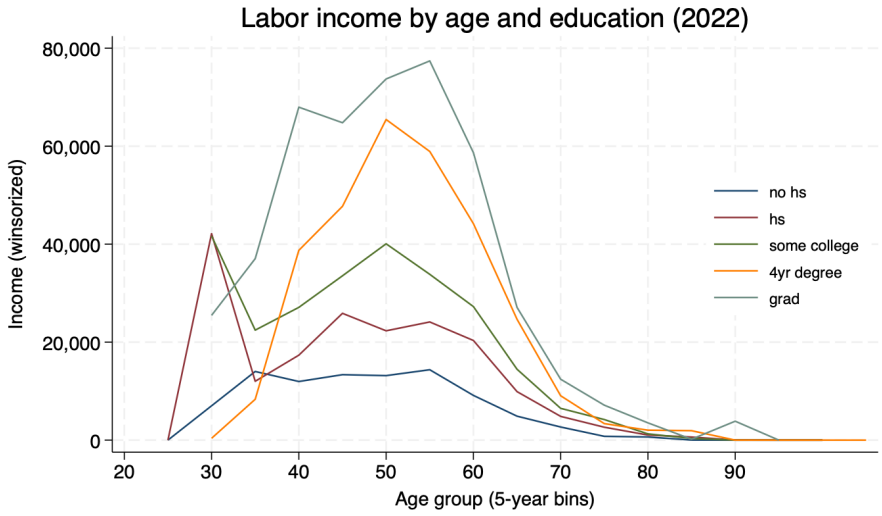

Figure 3: Labor income by age and education (2022)

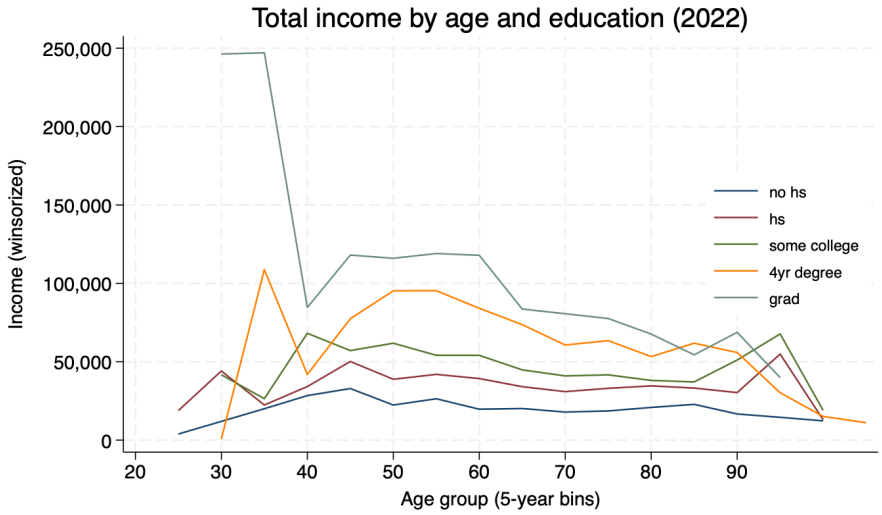

Figure 4: Total income by age and education (2022)

3.2 Wealth and asset class definitions

The HRS asks a number of questions aimed at measuring household wealth in the sample. I collect data on **interest income and dividends**, **capital gains**, **net investment flows**, and **previous period wealth holdings**, all of which are necessary to define the desired measure of household

returns. Since population-level administration data on these objects for individual taxpayers is not available in the U.S., it is important to understand each component needed in the return calculation. This will help to determine whether or not our measured distribution of returns in this dataset is sensible.

3.2.1 Capital gains

The possible asset classes for which capital gains can be computed in the HRS are i) primary residences, ii) secondary residences, iii) other real estate, iv) private business, v) IRA/Keogh (or "retirement"), vi) stocks/mutual funds, vii) bonds, viii) checking/savings/money market, ix) cds/t-bills, x) vehicles, xi) other assets. The possible liabilities are i) mortgages on primary residence, ii) mortgage on secondary residence, iii) other home loans, and iv) total other debt.

Capital gains for each asset class are computed as the estimated change in valuation across survey waves. The general trend is that, these changes in valuation tend to move around alot over the period. That said, capital gains are generally higher at the end of the period than at the start of the period for most asset classes. This can be seen for capital gains to busines ownership, in figure

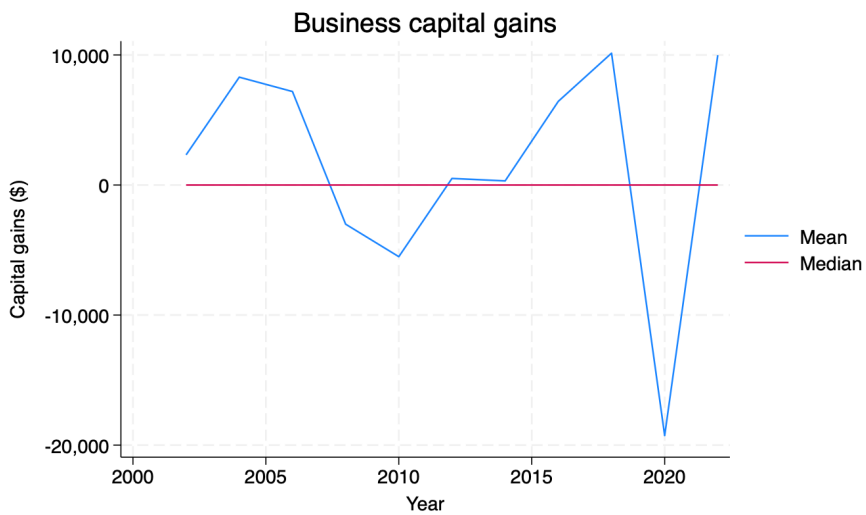

Figure 5: Capital gains: business, by year

The opposite is true for the bonds asset class: capital gains have generally fallen on average over the time period. This can be seen in the figure

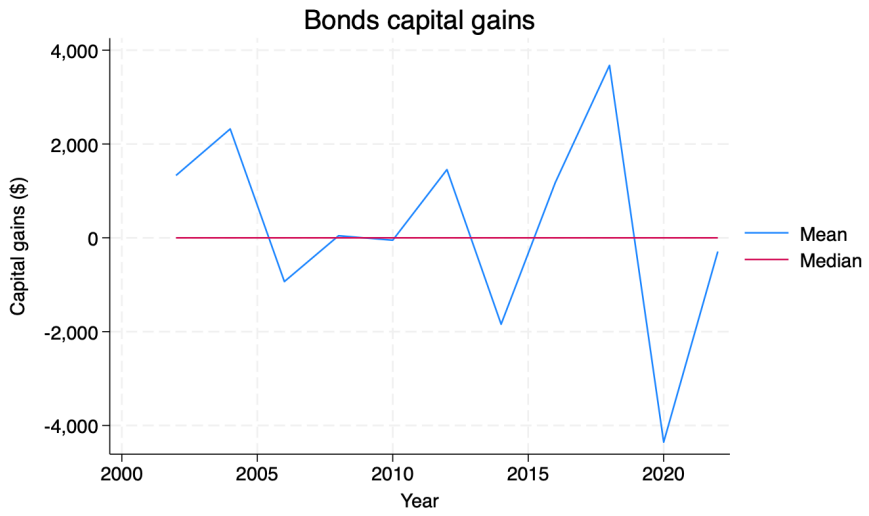

Figure 6: Capital gains: business, by year

Another feature that is apparent by taking a look at capital gains is that, most respondent hold no assets in a particular class. This can be seen by the vertical red line capturing a median of 0 for each wave of the survey. Residential assets generally are the bulk of individual portfolios, and yet the median is still 0 here in almost every wave. This is another sign that the data is sensible so far – large amount of non-participation despite evidence of returns is consistent with the empirically documented equity premium puzzle.

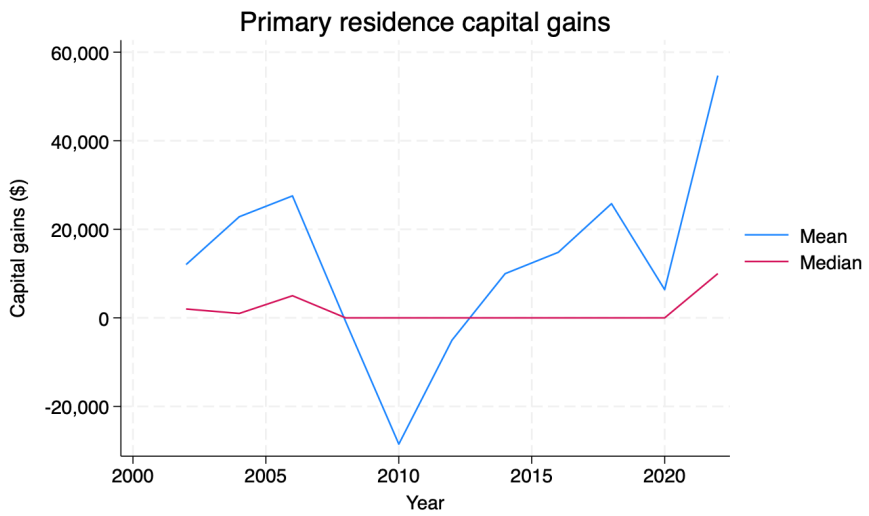

Figure 7: Capital gains: residential, by year

Notably, the dip in capital gain leading up to 2010 for primary residences, secondary residences, and real estate is also a good sign regarding how realibly measured the data is.

Figure 8: Capital gains: residential, by year

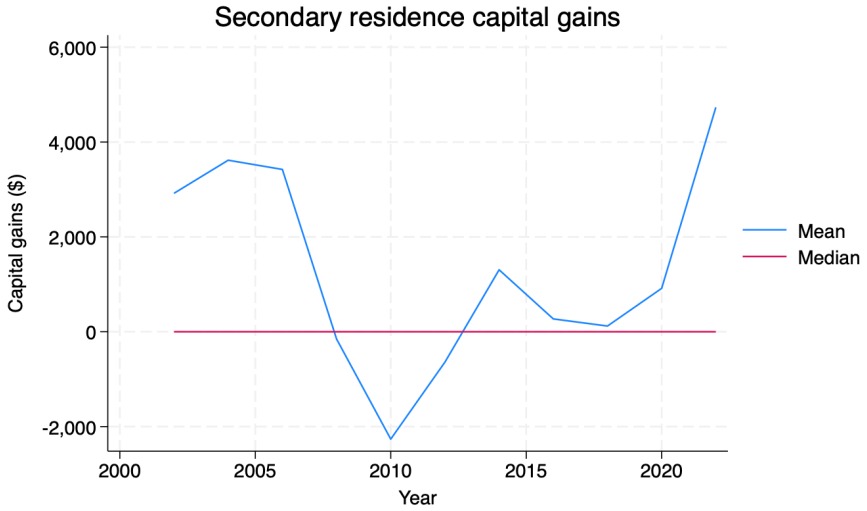

Figure 9: Capital gains: residential, by year

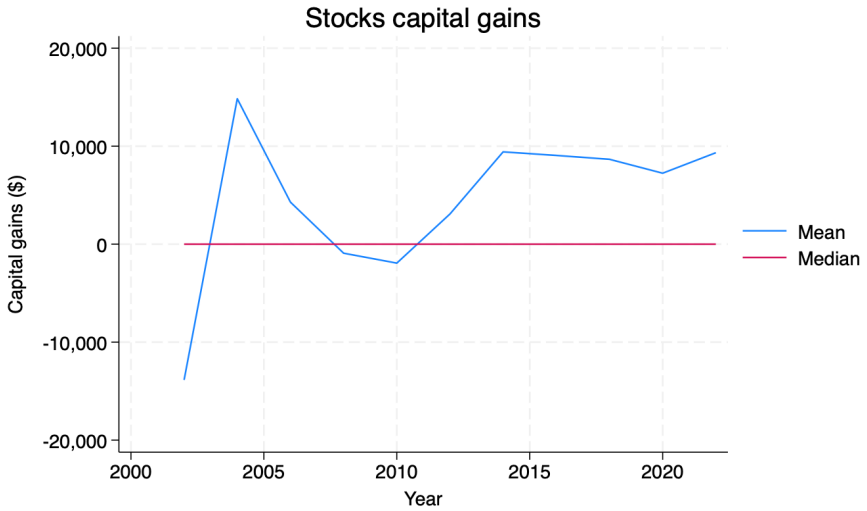

Figure 10: Capital gains: stocks, by year

3.2.2 Interest income and dividends

The HRS RAND longitudinal file has a distinctive way in which interest income and dividends received on these assets is measured. In particular, there is a variable which measures "household capital income received over the last calendar year, including business or farm income self-employment earnings, gross rent, dividend and interest income, trust funds or royalties, and other asset income". This particular feature of the data is the driving factor for the key modeling assumptions of this paper regarding measuring returns: asset classes are defined as narrowly as they can be given observation of interest income and dividends on those assets. For example, although we see capital gains for stocks, business, bonds, and real estate, we do not observe interest income or dividends on these assets individually. Thus, the narrowest asset class defined in this paper will be called "core" and will be comprised of these assets.

The RAND version of the data offers a number of variables measuring pension and annuity income as well as other forms of retirement income. I use this to construct a measure of returns to retirement assets and to define a broader notion of portfolio returns by considering returns to core and retirement assets. A key assumption is that there are no interest income or dividends earned on residential assets[2]. With this assumption, we can consider interest income and dividends on the entire portfolio as the same as interest income and dividends on the portfolio with just core and retirement assets. Thus, to compute a measure of returns to net wealth, I need to add in the remaining available capital gains per asset class (for those that receive no interest income).

[2]Or on any of the asset classes for which capital gains are available for.

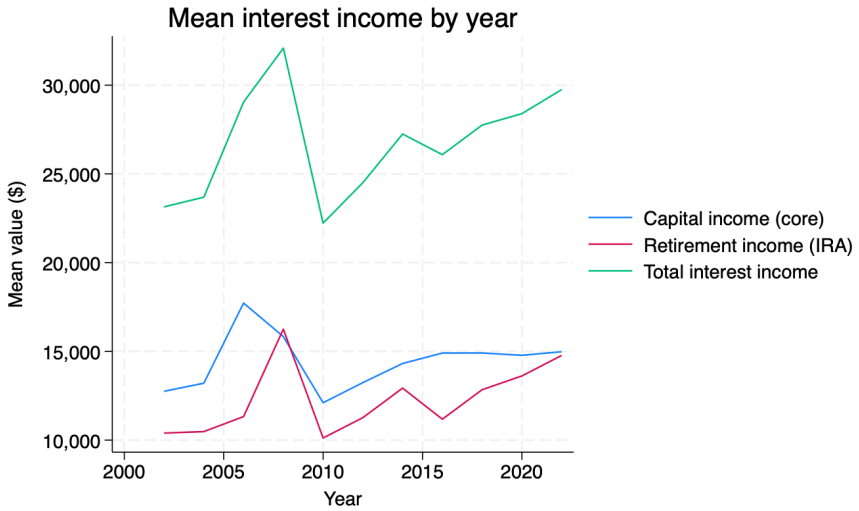

Figure 11: Interest income mean by year

3.2.3 Net investment flows

Net investment flows are vital in computing returns accurately because capital gains and interest income can miss other flows of value into a given asset class. In the HRS dataset, net investment flows are the only variable relevant to the returns calculation that RAND did not clean and process. For this reason, there are substantially less observation for this variable per asset class than the others. To work around this, I do two things. First, I process the net investment flow per asset class myself by using the associated flag variable (asking yes or no if an individual has "bought or sold since the previous wave") for each variable. Second, I assume that if an individual receives interest income on that asset class, but their flow in nonmissing, than the nonmissing flow to that asset class is treated as 0.[3]

With this in mind, I present a figure of net investment flows into the assets available in the dataset. As you can see, the magnitudes for flows into a given class are comparable. I use these flows to construct net investment flows into i) core assets, ii) retirement assets, iii) residential assets, iv) core+residential assets, and v) net wealth.

[3]Non-missing interest income (and capital gains) indicate participation within an asset class.

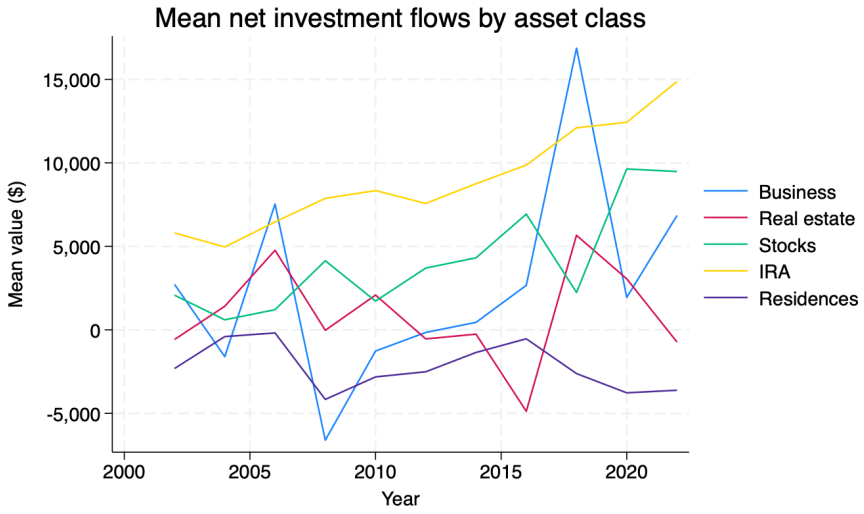

Figure 12: Flows by asset, mean by year

3.2.4 Wealth and inequality

After defining asset classes, I consider mean wealth within these narrowly defined asset classes for each year of the sample. Interestingly, retirement assets seem to perform the worst in terms of average returns. Core assets seem to offer the best performance, and all asset classes see a downward dive leading up to 2010 – historically accurate in the context of investment performance during the global financial crisis.

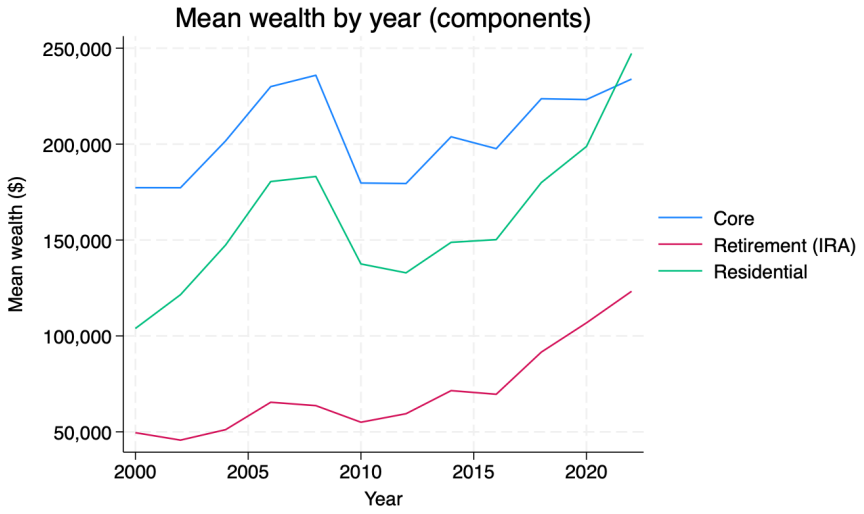

Figure 13: Wealth mean by year (components)

I turn attention to measures of wealth in the sample based on the portfolio definitions: i) net wealth in core assets, ii) net wealth in core and retirement assets, iii) total net wealth.

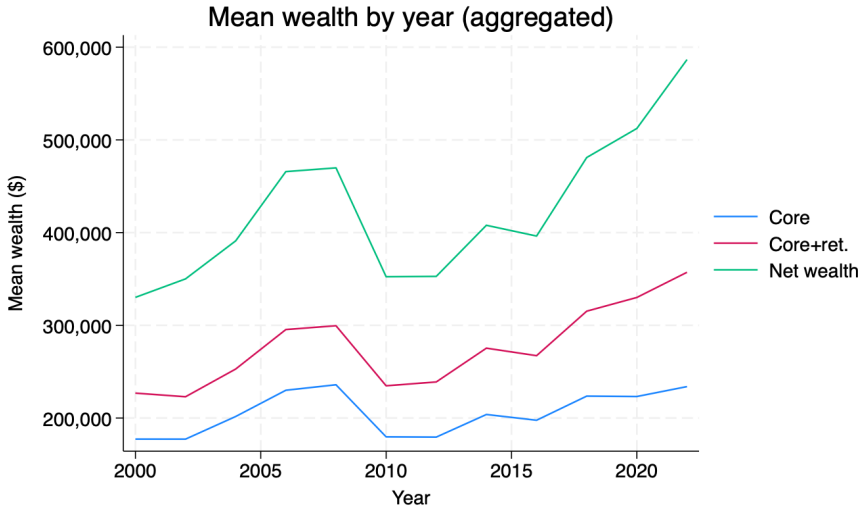

Figure 14: Wealth mean by year (aggregated)

I consider the distribution of these measures of income and wealth across respondents of the survey in the following figures. For income, the measure of labor income is significantly more unequally distributed than the measure of total income in the sample. This is likely an artifact

of the sampling method of the HRS (older likely overrepresented and are towards the end of their working years).

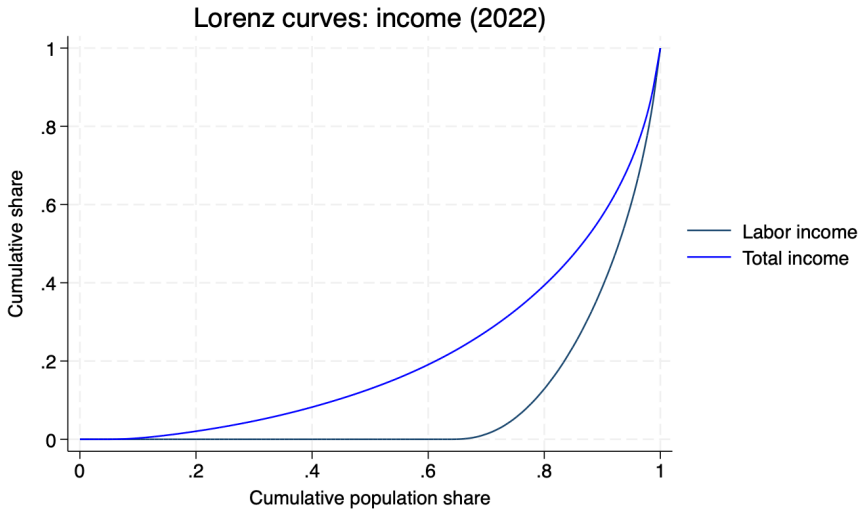

Figure 15: Lorenz: income (2022)

For wealth, holdings in core assets and retirement assets are more unequally distributed than holdings in residential assets. This is in line with common knowledge that home ownership is the most common form of asset ownership in the U.S.

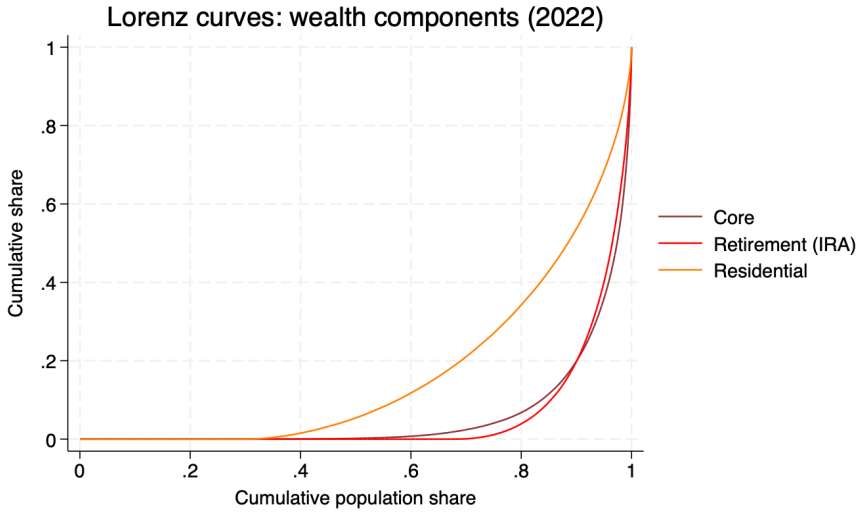

Figure 16: Lorenz: wealth components (2022)

Interestingly, the distribution of net wealth become less unequal when it is extended to incorporate retirement assets on top of core assets. this can be seen in the following figure .

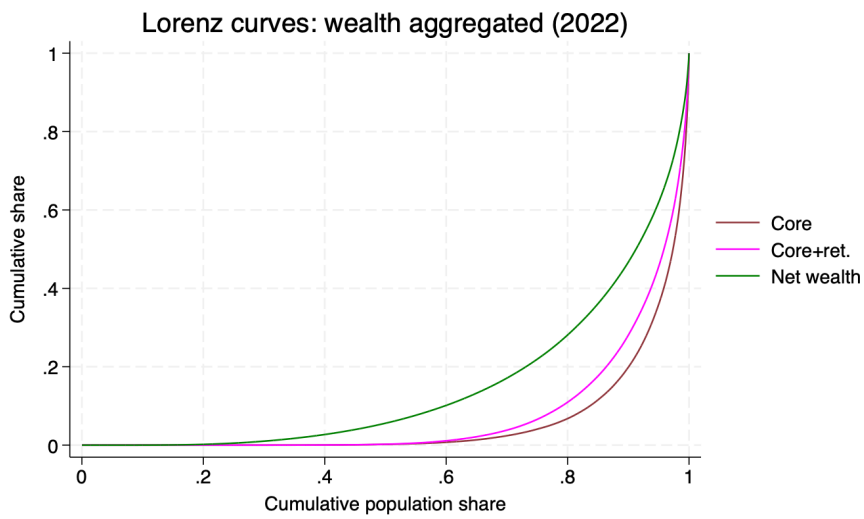

Figure 17: Lorenz: wealth aggregated (2022)

3.2.5 Portfolio composition

To describe the composition of portfolio during the period, I document mean portfolio shares for the relevant asset classes.

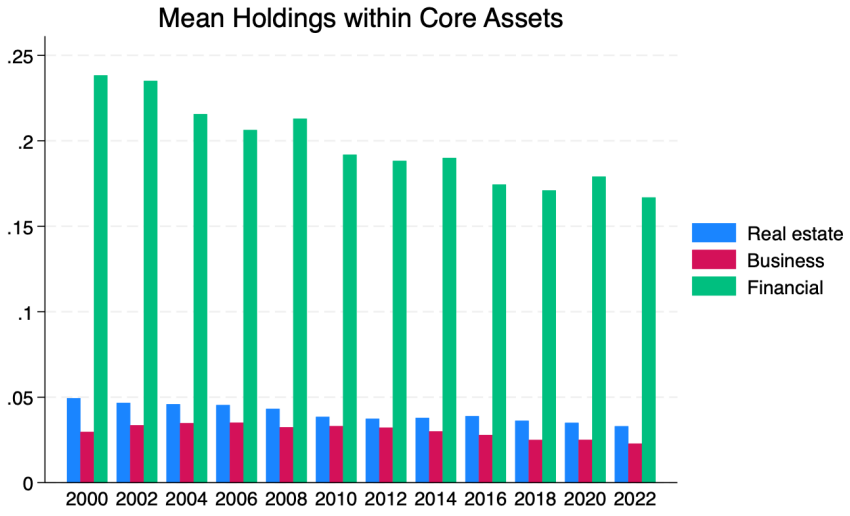

Figure 18: Mean holdings within Core assets, by year

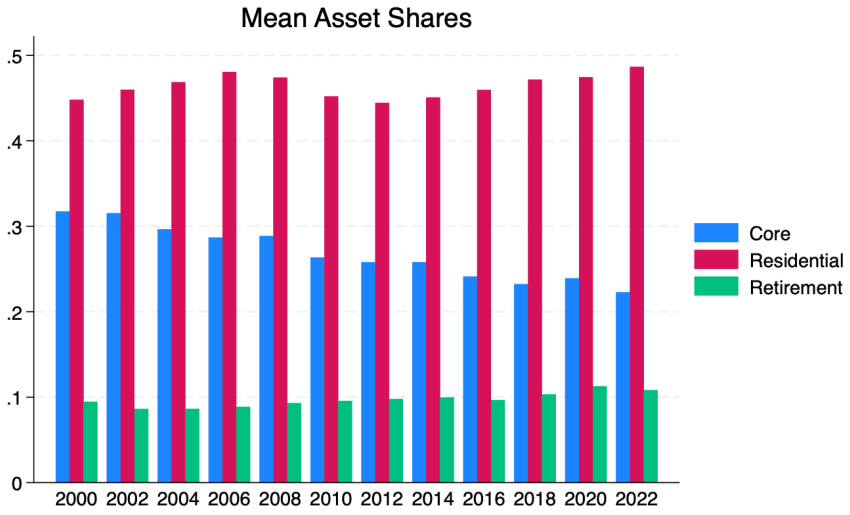

Figure 19: Mean Asset shares, by year

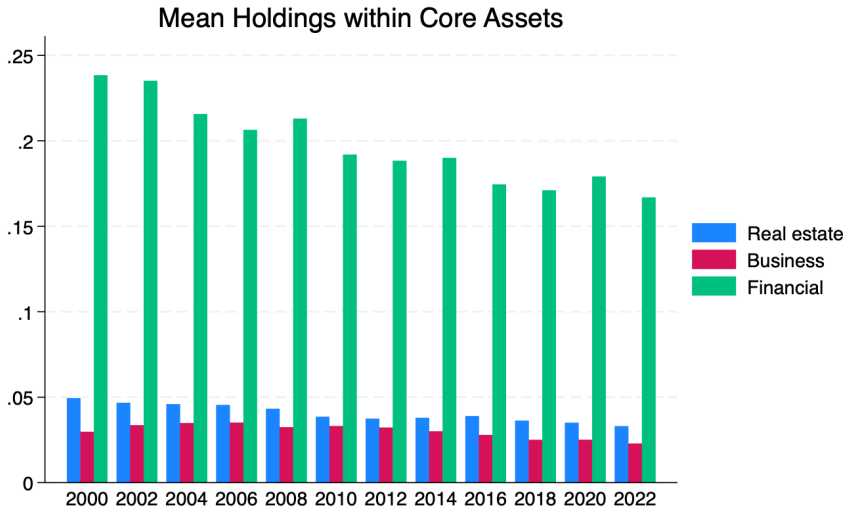

Figure 20: Mean holdings within Liabilities, by year

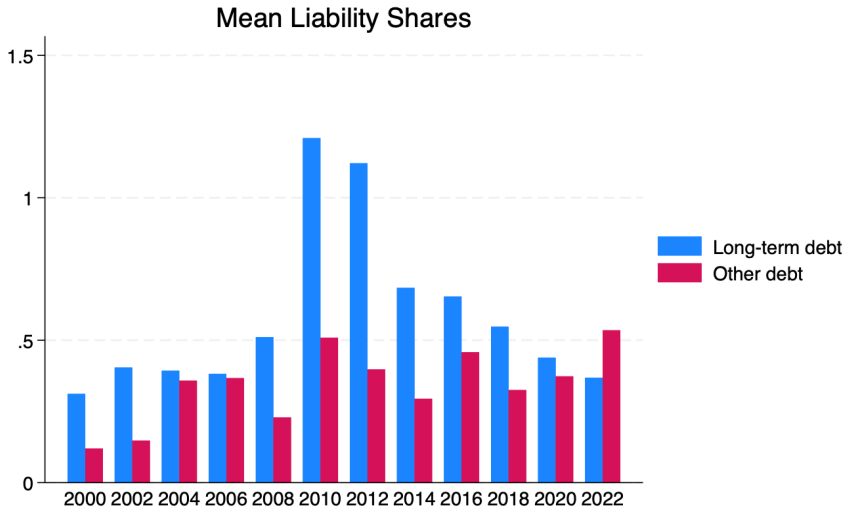

Figure 21: Mean Liability shares, by year

Investment behavior is likely to vary with income and wealth. Asset shares (core, residential, retirement) by percentile of labor income, total income, gross wealth, and net wealth (2022):

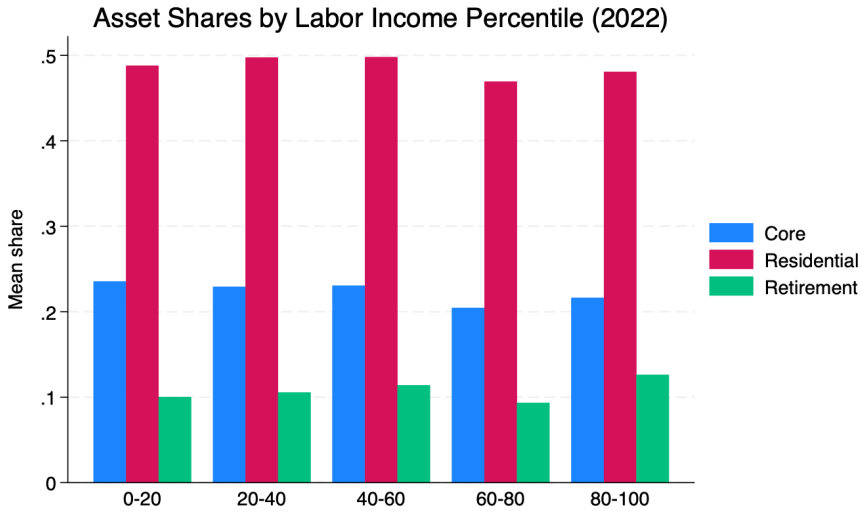

Figure 22: Asset shares by labor income percentile (2022)

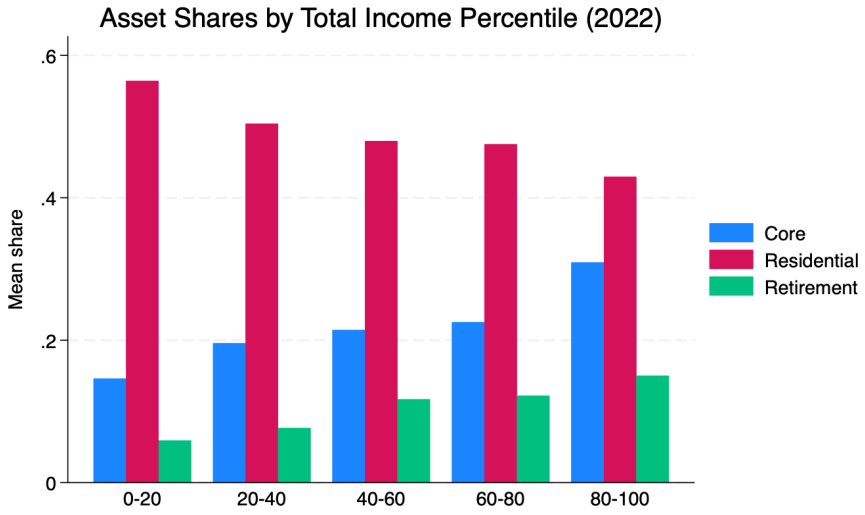

Figure 23: Asset shares by total income percentile (2022)

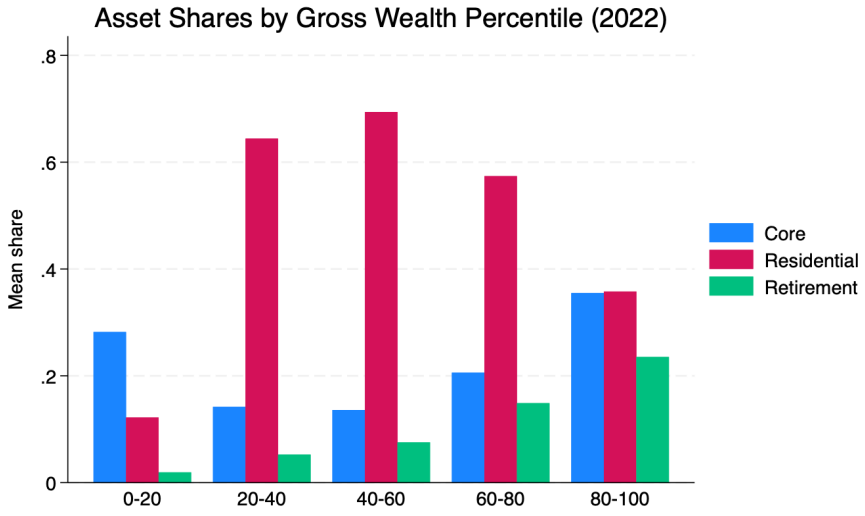

Figure 24: Asset shares by gross wealth percentile (2022)

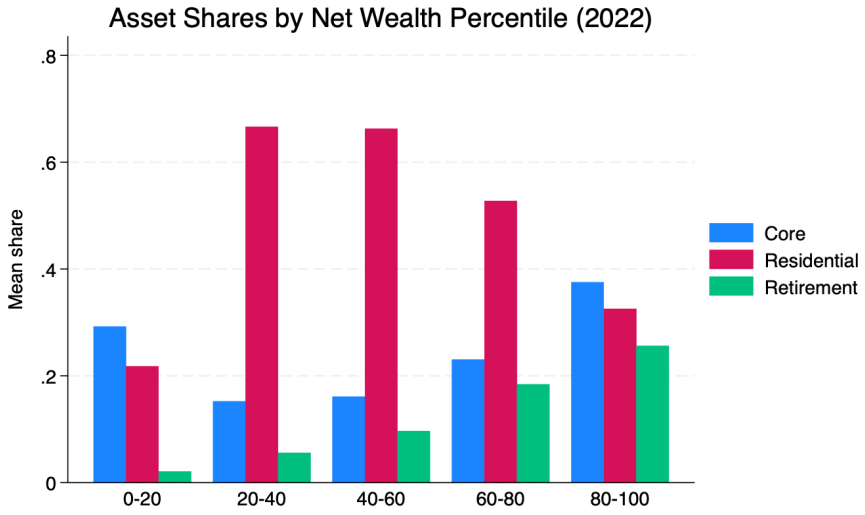

Figure 25: Asset shares by net wealth percentile (2022)

We can also look at the same breakdown by income and wealth percentiles but for the ratio of debt to gross assets. This is captured in the following figures .

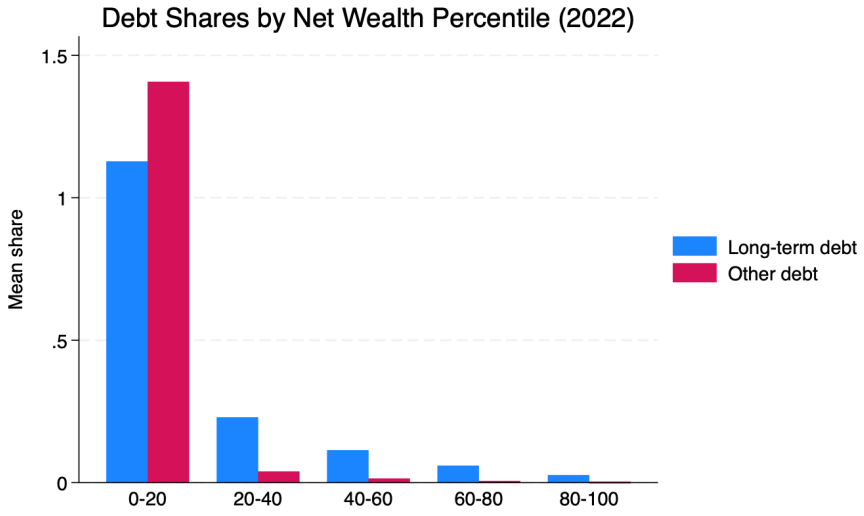

Figure 26: Debt shares by labor income percentile (2022)

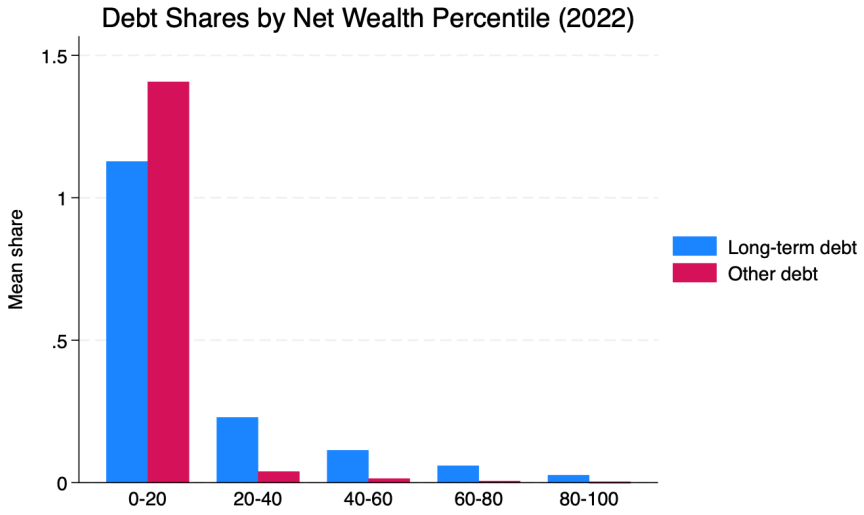

Figure 27: Debt shares by total income percentile (2022)

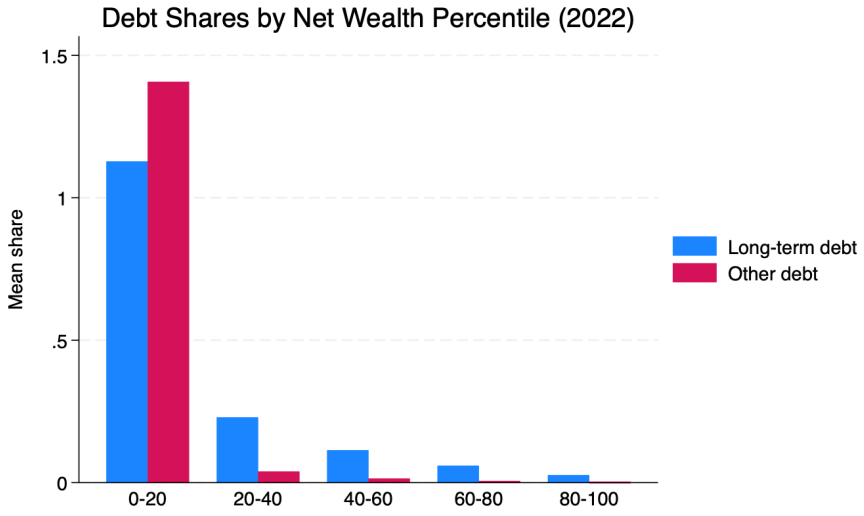

Figure 28: Debt shares by gross wealth percentile (2022)

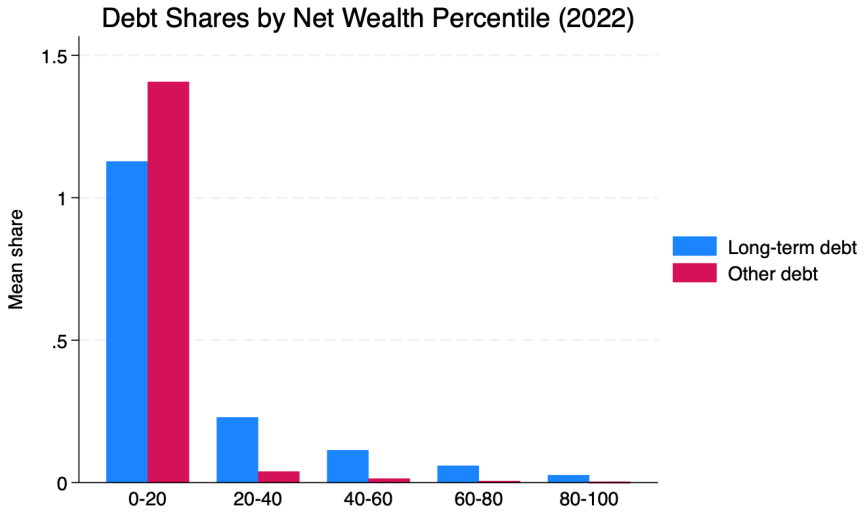

Figure 29: Debt shares by net wealth percentile (2022)

3.3 Returns

I follow recent literature using Norwegian administrative tax data at the population level and and using PSID in terms of the components used to construct a measure of returns. However, I take seriously the structure of the dataset I am working with(the HRS RAND longitudinal file) in

defining the portfolio/asset class for which the returns are being accrued to. As mentioned before, core assets is the most narrowly defined asset class here because interest income and dividends does not disaggregate into the asset classes: stocks, bonds, private business, IRA/Keogh.

The following formula from Daminato and Pistaferri 2024 serves as motivation for the return measures defined using the HRS data:

$$r_t = \frac{y_t^c + cg_t - F_t - y_t^d}{A_{t-1} + .5F_t}$$

where y_t^c interest income and dividends, capital gains cg_t measured as the difference between reported stock across waves, F_t net investment flows, y_t^d payments on debt (in the RAND longitudinal file, the variables were mentioned are all in net terms so this variable was 0), and A_{t-1} total net wealth at beginning of previous period.

3.3.1 Narrow asset classes

Recall that we only see interest income and dividends for core assets and retirement assets. Since this is an important component of the returns calculation, we construct the narrowest return measure on core assets. To do so, I keep track of all asset classes captured in the capital gains variable[4] and aggregate the capital gains and net investment flows for this group of assets.

Returns to core assets are given by

$$r_{core,t} = \frac{y_{core,t}^c + cg_{core,t} - F_{core,t}}{A_{core,t-1} + .5F_{core,t}}$$

where $y_{core,t}^c$ is the interest income and dividents on the core assets, $cg_{core,t}$ are the capital gains on the core assets[5], $F_{core,t}$ are the net investment flows on core assets, and importantly, $A_{core,t-1}$ is the total value of the core assets in the previous period.

The rest of the return measures are defined similarly. In the HRS data, we can see income on retirement assets as well. This allows us to define a return measure for retirement assets:

$$r_{ret.,t} = \frac{y_{ret.,t}^c + cg_{ret.,t} - F_{ret.,t}}{A_{ret.,t-1} + .5F_{ret.,t}}$$

where $y_{ret.,t}^c$ is the interest income and dividents on the retirement assets, $cg_{ret.,t}$ are the capital gains on the retirement assets, $F_{ret.,t}$ are the net investment flows on retirement assets, and $A_{ret.,t-1}$ is the total value of the retirement assets in the previous period.

Based on the modules offered in a given wave of the survey, assume that there are no interest income or dividends to either primary and secondary residential assets, so that the interest income and dividends accrued to residential assets is 0 in a given period ($y_{ret.,t}^c$). With this in mind, returns to residential assets are defined as:

$$r_{res.,t} = \frac{cg_{res.,t} - F_{res.,t}}{A_{res.,t-1} + .5F_{res.,t}}$$

[4]Those asset classes are: other real estate, business/farm, stocks/mutual funds, bonds, checking/savings, and CDs/t-bills.

[5]where the capital gains for each asset class is $V_{i,t} - V_{i,t-1}$, where $V_{i,t}$ is the value of the asset (business or stock or real estate) for the i-th respondent in the t-th wave of the survey.

where $cg_{res.,t}$ are the capital gains on the residential assets, $F_{res.,t}$ are the net investment flows on residential assets (including improvement costs), and $A_{res.,t-1}$ is the total value of the residential assets in the previous period.

3.3.2 Broad asset classes

After defining returns for narrow asset classes (i.e. returns to a single asset), I also define return measure for portfolios (returns on multiple assets). The first is for returns to both core and retirement assets:

$$r_{coret.,t} = \frac{y^c_{coret.,t} + cg_{coret.,t} - F_{coret.,t}}{A_{coret.,t-1} + .5F_{coret.,t}}$$

where $y^c_{coret.,t}$ is the interest income and dividends on core and retirement assets, $cg_{coret.,t}$ are the capital gains on the retirement assets, $F_{coret.,t}$ are the net investment flows on retirement assets, and $A_{ret.,t-1}$ is the total value of the retirement assets in the previous period.

Returns to net wealth is given by

$$r_{net,t} = \frac{y^c_{net.,t} + cg_{net.,t} - F_{net.,t}}{A_{net.,t-1} + .5F_{net.,t}}$$

where $y^c_{net.,t}$ is the interest income and dividends on core and retirement assets, $cg_{net.,t}$ are the capital gains on all available asset classes in the survey, $F_{net.,t}$ are the net investment flows on assets for which it is relevant in the survey (stocks, bonds, business, IRA, residences, and real estate), and $A_{net.,t-1}$ is the total value of the available assets minus liabilities in the previous period[6].

I present the means for the return measures in the following figures. I group them by the returns at the asset level (core, retirement, residential) and at the portfolio level (core, core and retirement, net wealth).

[6] Mortgages on secondary residences are left out of the list of liabilities since it was an optional question in some waves of the survey.

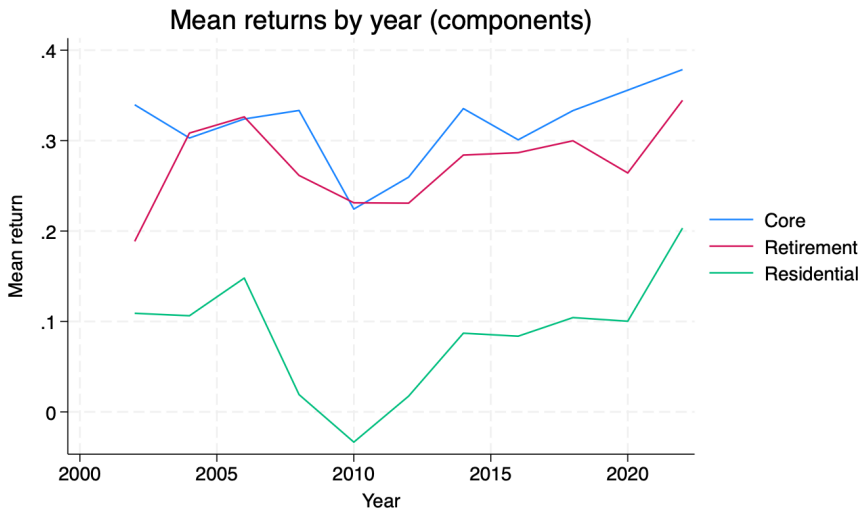

Figure 30: Returns mean by year (components)

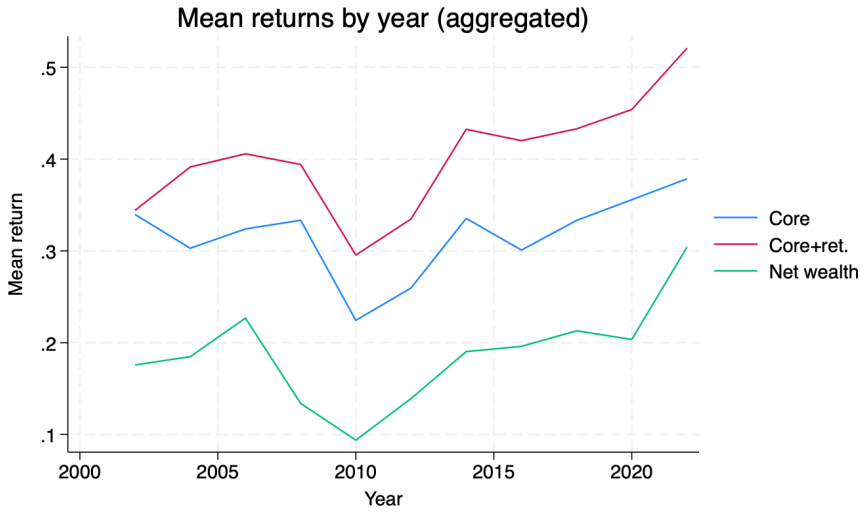

Figure 31: Returns mean by year (aggregated)

The following figures give a better idea of the distribution of the return measures across respondents.

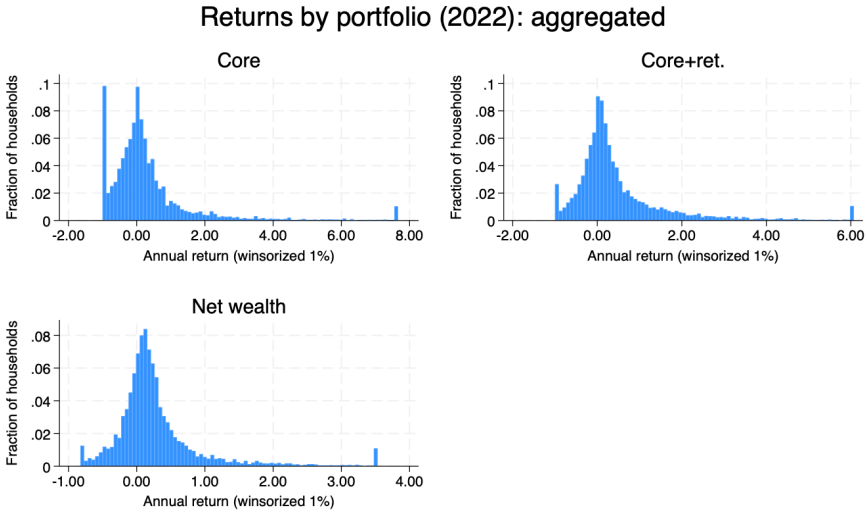

Figure 32: Returns histogram, aggregated (2022)

These are encouraging, as shape of this distribution closely resembles early estimates of the empirical distribution of individual realized returns in the Norwegian population data.

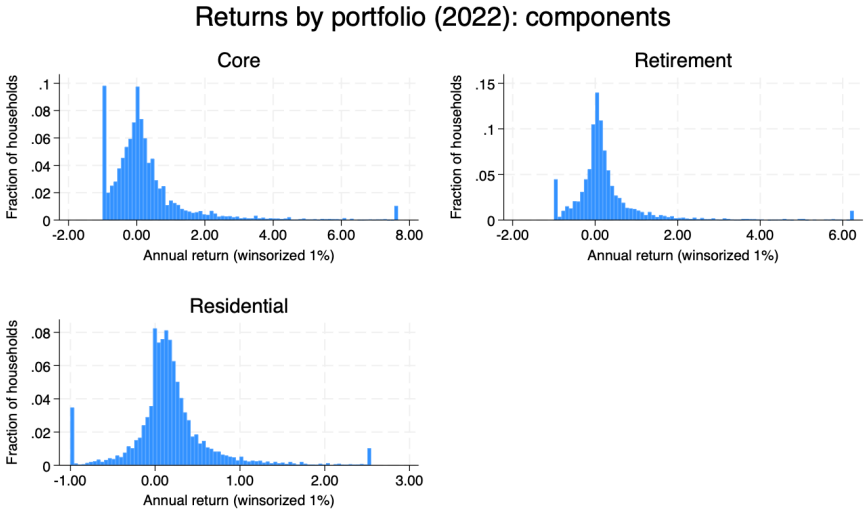

Figure 33: Returns histogram, components (2022)

3.4 Income and wealth

Income and wealth are understood to be positively correlated. To see this play out in the HRS data, I first look at mean labor income by wealth percentile. In ths figure, it is clear that a positive relationship seems to hold.

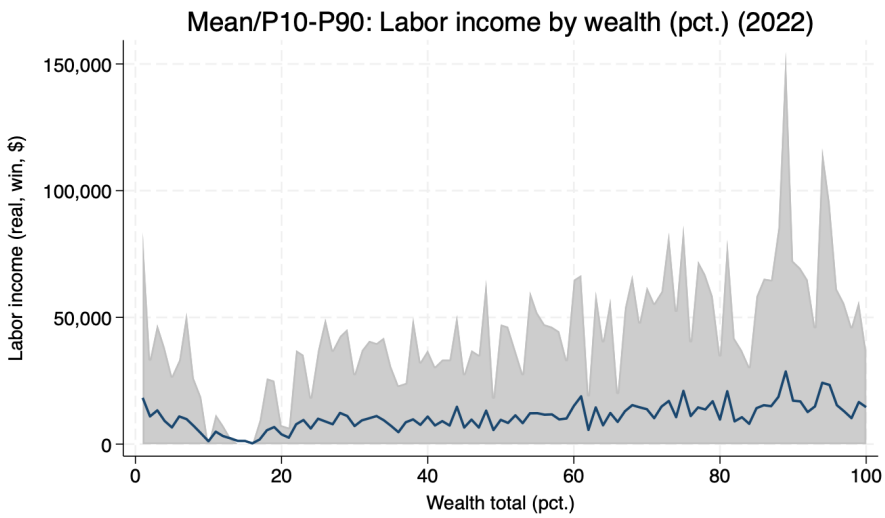

Figure 34: Labor income mean/P10-P90 by wealth percentile (2022)

The literature on income and wealth also documents significantly more inequality in the the upper tails of the wealth distribution than in the income distribution. The observations towards to top and bottom of the wealth distribution in the following figure seem to suggest a non-linear relationship between the variables.

Figure 35: Log labor income binscatter (2022)

I repeat this for the measure of total income and both patterns persist.

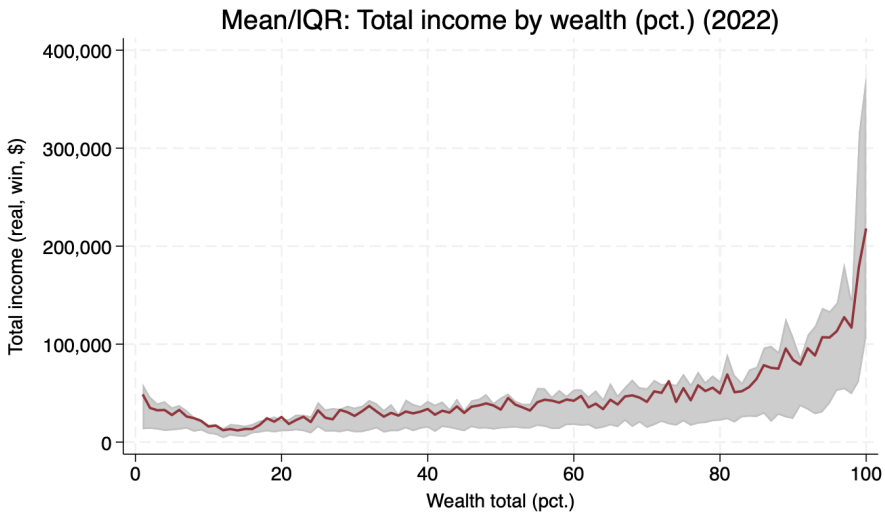

Figure 36: Total income mean/IQR by wealth percentile (2022)

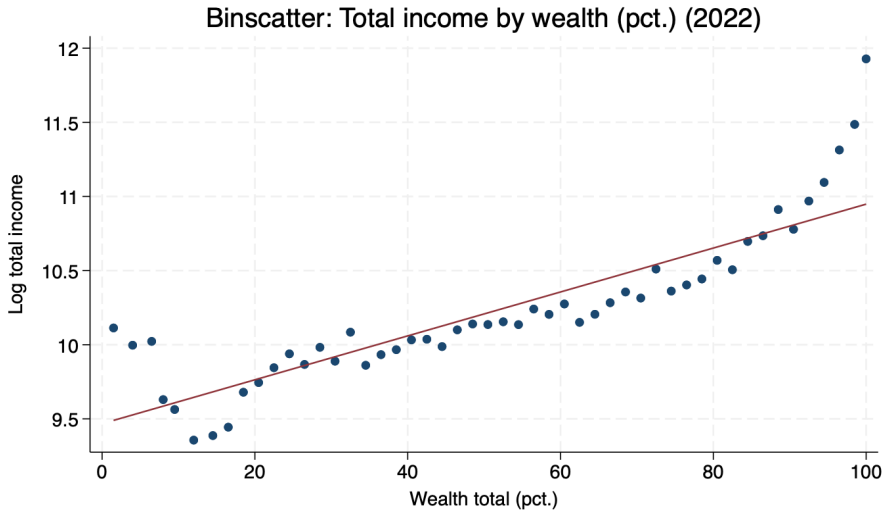

Figure 37: Log total income binscatter (2022)

3.5 Returns and wealth by portfolio

A positive relationship between wealth and returns has also been documented, referred to as *scale dependence*. In the next three figures, I show that there does seem to be a positive statistical relationship between the return measure (core, retirement, residential) and wealth.

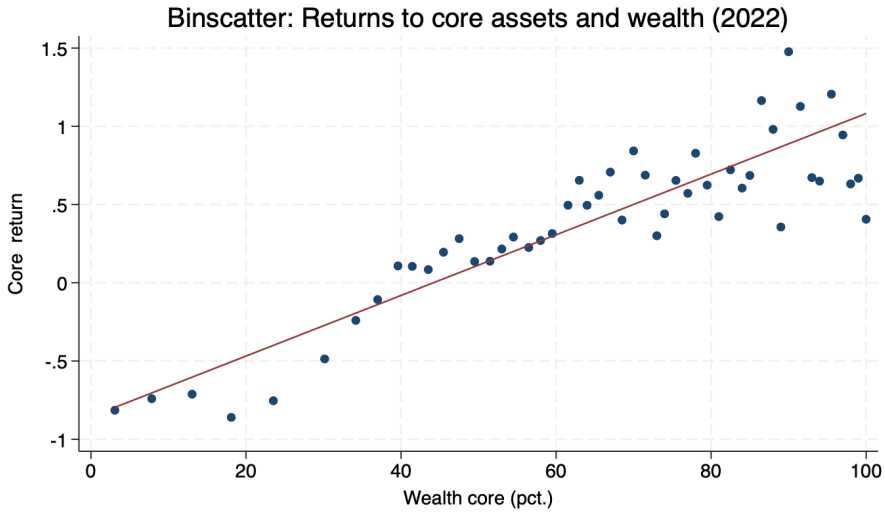

Figure 38: Core return binscatter (2022)

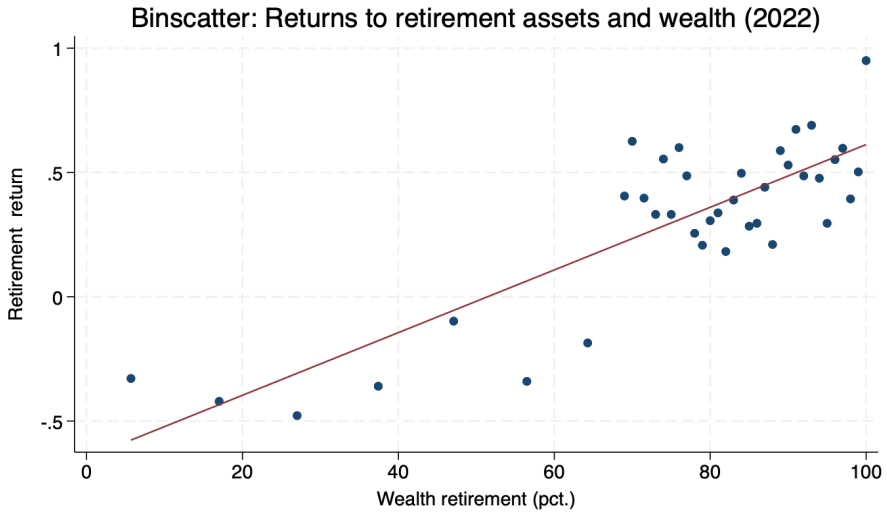

Figure 39: Retirement return binscatter (2022)

Figure 40: Residential return binscatter (2022)

I also show the positive statistical relationship between returns and wealth at the portfolio level, when comprised of core assets and retirement assets.

Figure 41: Core+IRA return binscatter (2022)

Interestingly, the relationship between wealth and returns to total net wealth does not appear positive. In fact, they correlation is slightly downward-sloping.

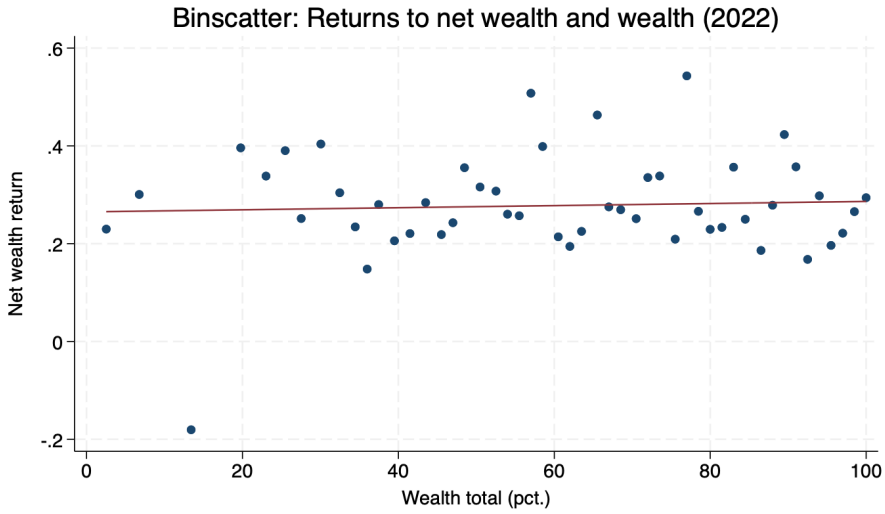

Figure 42: Net wealth return binscatter (2022)

3.6 Trust

I turn my attention to the suite of trust questions in "Section V: Modules" section of the 2020 HRS data. There are a total of 8 questions, asking respondents to say on a scale of 1-10 "how much do you trust people in general?" and of their trust in other features of American life relating to healthcare, finance, and media. These questions were only asked for a single year. Although this is an issue, because the panel structure of the HRS allows me to estimate the persistent component of returns, which is an important object in this literature. That said, in the pooled setting, we can assume that trust, like education, is fixed over time.[7]

I turn my attention to the correlations between the trust variables in Table 3.6. There is significant correlation between the trust measures.

Table 5: Trust variables correlation matrix

	General trust	Social Security	Medicare	Banks	Financial advisors	Mutual funds	Insurance	Media
General trust	1.00	0.30	0.26	0.37	0.33	0.36	0.39	0.21
Social Security	0.30	1.00	0.84	0.45	0.33	0.26	0.43	0.29
Medicare	0.26	0.84	1.00	0.41	0.31	0.26	0.38	0.27
Banks	0.37	0.45	0.41	1.00	0.55	0.43	0.46	0.27
Financial advisors	0.33	0.33	0.31	0.55	1.00	0.63	0.50	0.26
Mutual funds	0.36	0.26	0.26	0.43	0.63	1.00	0.41	0.26
Insurance	0.39	0.43	0.38	0.46	0.50	0.41	1.00	0.32
Media	0.21	0.29	0.27	0.27	0.26	0.26	0.32	1.00

Pairwise correlations between trust items (2020).

From here, I perform a principal component analysis and store the first two components for use in the statistical analysis. The loadings on each of the trust measures is given in Table 3.6.

[7] Literature on trust talks about how history of being cheated or treated fairly form individuals' trust over time. If this is true and at some point, one learns enough an forms their trust level, then a sample with an overrepresentation of older household is a reasonable environment to assume constant trust levels.

Table 6: Trust PCA loadings (first two components)

Trust item	PC1	PC2
General trust	0.3002	0.1808
Social Security	0.3835	-0.5451
Medicare	0.3657	-0.5680
Banks	0.3887	0.1014
Financial advisors	0.3844	0.3757
Mutual funds	0.3480	0.4309
Insurance	0.3786	0.0971
Media	0.2565	-0.0317

Principal components on trust variables (2020).

I am interested in the relationship between trust and economic performance as the literature is, however in my setting the measure of economic performance is the return to assets. If a hump-shaped relationship is reasonable for income, it is even more plausible for returns: there is an inherent (and possibly explicit) level of trust between borrower and lender when forming credit contracts.

That said, it is important to understand the nature of the trust measure in the HRS sample. I begin to do this by considering each trust measure conditional on race/ethnicity. There seems to be significant group variation in means.

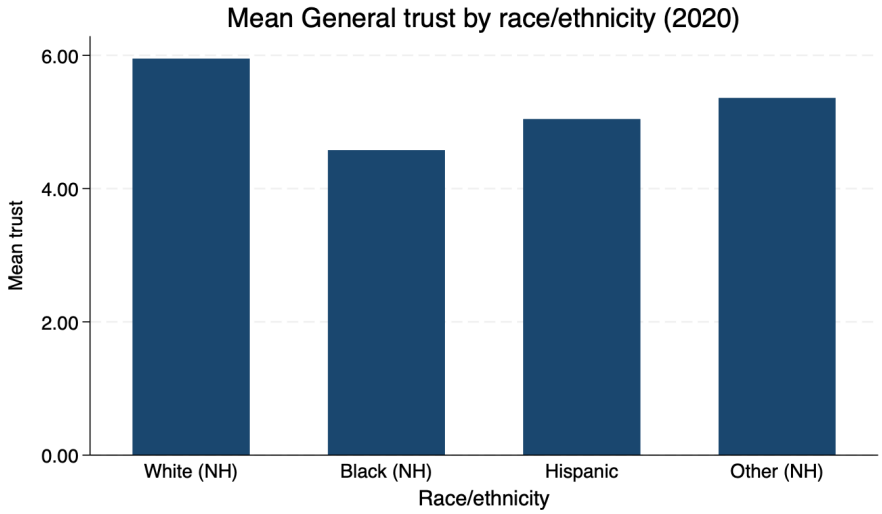

Figure 43: Mean general trust by race/ethnicity (2020)

3.6.1 Trust and income

As discussed before, the literature on trust and economic performance suggests there's a "right amount of trust". I want to see if this hump shaped relationship holds between the measures of trust and income in the HRS. After deflating and, more importantly, winsorizing labor income at the top and bottom 1%, the relationship seems to be hump shaped.

Income in the HRS is highly right-skewed and features a large mass at zero labor income (reflecting retirement and non-participation). In applied work, a common baseline is to use log(income) because it reduces the influence of extreme values and yields coefficients interpretable as semi-elasticities. However, $\log(\cdot)$ is undefined at zero (and problematic with negative values when they arise from measurement or netting conventions), so using log(income) in the HRS mechanically drops a large fraction of observations and can substantially change the visual and econometric relationship.

In our data, the corresponding scatter of log(income) versus trust changes markedly because so many observations have exactly zero labor income. Using $\log(1+\text{income})$ retains zeros but imposes an ad hoc curvature near zero and, in our case, does not restore the same qualitative picture.

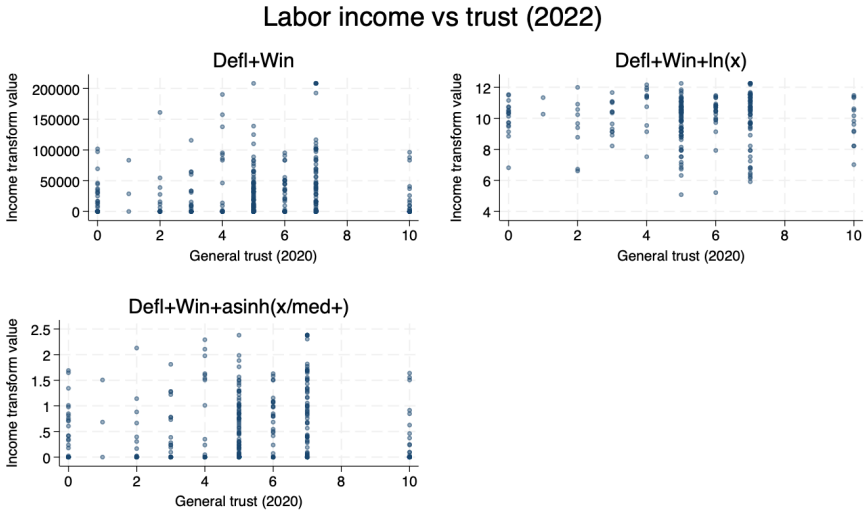

Figure 44: Labor income measures vs. trust (2022)

Motivated by this issue, we follow a large applied econometrics literature that uses the inverse hyperbolic sine (IHS, or asinh) transformation as a "log-like" mapping that is well-defined at zero and can accommodate negative values. The IHS transformation was advocated early on as an alternative to the log for heavy-tailed outcomes with extreme values and potential zeros/negatives (Burbidge, Magee, and Robb, *"Alternative Transformations to Handle Extreme Values of the Dependent Variable"*; MacKinnon and Magee, *"Transforming the Dependent Variable in Regression Models"*). More recently, Bellemare and Wichman (*"Elasticities and the Inverse Hyperbolic Sine Transformation"*) provide practical guidance on interpretation, emphasizing that $\text{asinh}(y)$ behaves similarly to $\log(y)$ for large y (since $\text{asinh}(y) \approx \log(2y)$ when y is large) while remaining defined at $y = 0$.

A related point is that the IHS can be sensitive to the units of measurement, which motivates a *scaled* IHS of the form $\text{asinh}(y/\kappa)$ (see discussions in Aïhounton and Henningsen, *"Units of Measurement and the Inverse Hyperbolic Sine Transformation"*; Norton, *"The Inverse Hyperbolic Sine Transformation"*). In practice, κ is chosen as a meaningful scale (e.g., the median positive income), which makes the transformation more comparable across samples and improves interpretability near zero. Empirically, applying the (scaled) IHS to our deflated and winsorized income restores a scatter with trust that is visually well-behaved while retaining the economically important mass at zero labor income. This can be seen for the measure of toal income as well in the following figure .

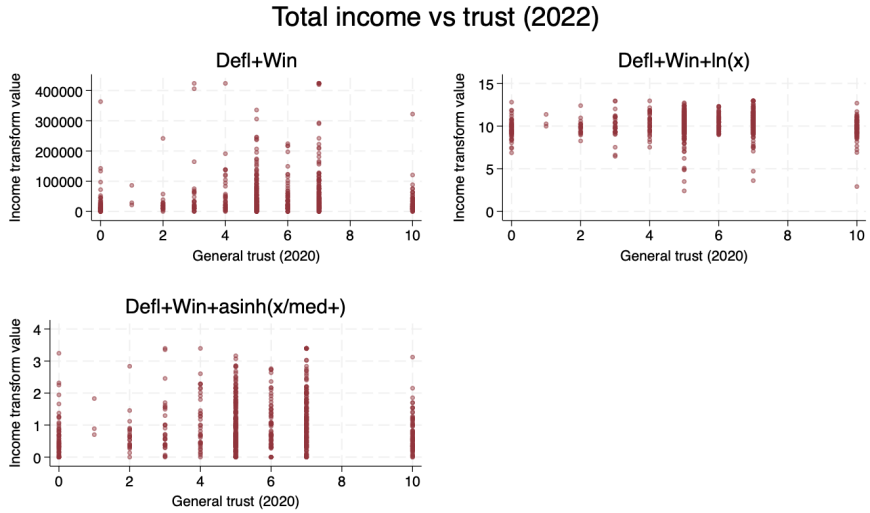

Figure 45: Total income measures vs. trust (2022)

3.6.2 Trust and returns

The scatterplots for trust and the measures of returns suggest stronger evidence for this hump shaped relationship. Especially for the smaller portfolio compositions (core, core and retirement), return values are highest in the middle and lower at the lowest and highest trust values.

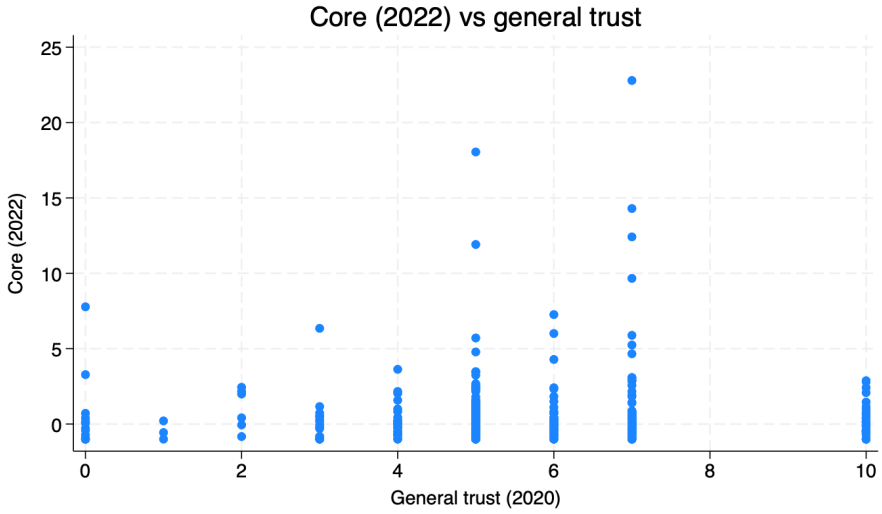

Figure 46: Returns to core assets vs. trust (2022)

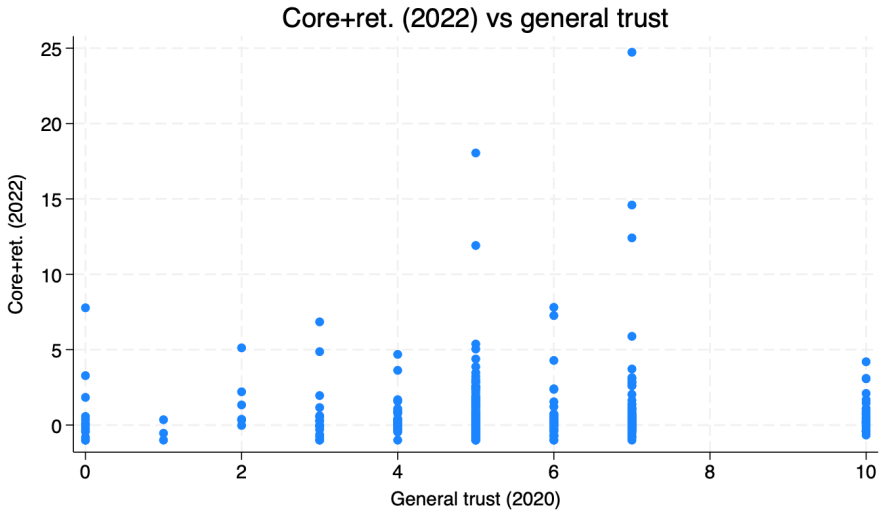

Figure 47: Returns to core and retirement assets vs. trust (2022)

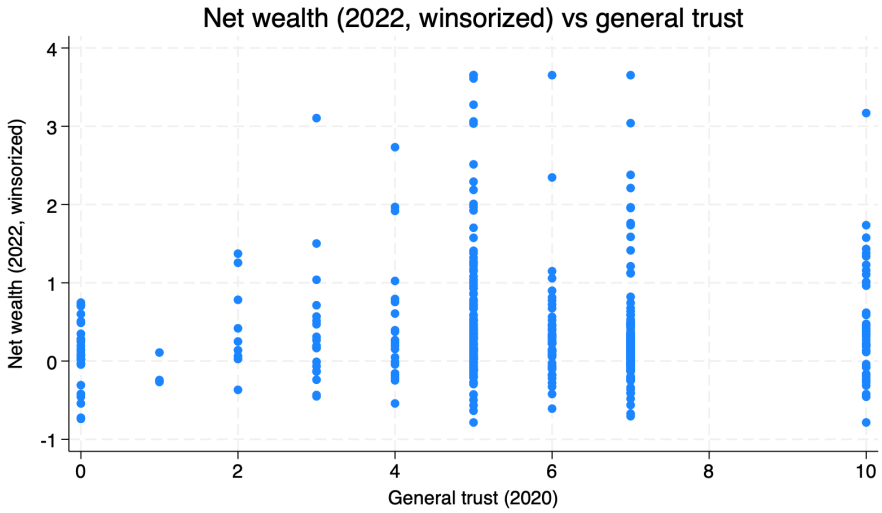

Figure 48: Returns to net wealth vs. trust (2022)

3.7 Demographics

As the HRS oversample older households, I take a quick look at demographic variables that will be important for the statistical analysis later.

Table 7: Demographics: general controls (2020)

Variable	N	Mean
Age	15723	68.089
Female	15723	0.594
Years of education	15651	12.961
Married	15685	0.537
Immigrant	15718	0.172
Born in U.S.	15718	0.828
Race: White (NH)	8865	0.565
Race: Black (NH)	3381	0.216
Race: Hispanic	2678	0.171
Race: Other (NH)	765	0.049
Working (in labor force)	15479	0.386

Table 8: Demographics: other controls (2020)

Variable	N	Mean
Depression	14998	1.55
Health conditions	15723	2.39
Medicare	15498	0.60
Medicaid	15383	0.14
Life insurance	15340	0.53
Times divorced	15723	0.57
Times widowed	15723	0.22

Mean = pct.

The literature on trust already documents a signficant statistical relationship between trust and mental health. For that reason, I use the HRS RAND measure for a mental health index and show a clearly negative, linear relationship with trust.

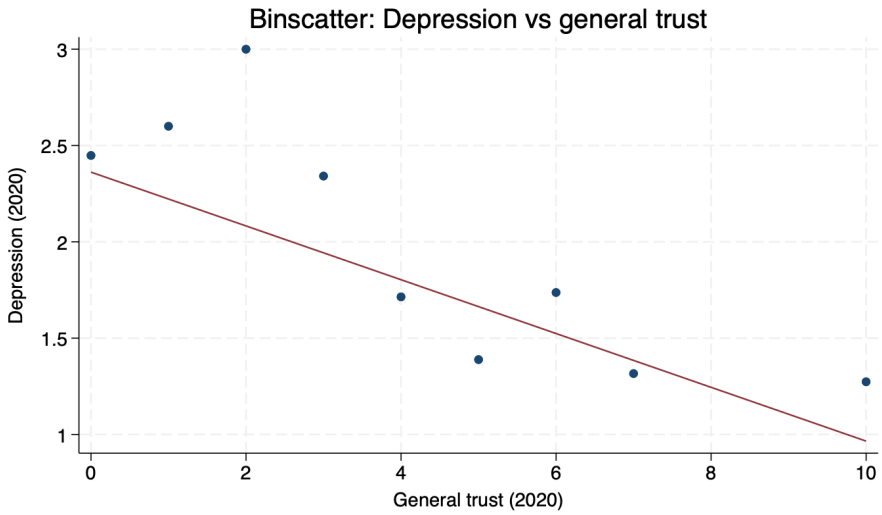

Figure 49: Depression vs. trust

4 Results

4.1 Determinants of trust

Table 4.1 shows the empirical models for trust examined in this paper. As Alesina and Ferrara 2000 suggests, demographic variables tend to determine trust. In the HRS data, the estimated coefficients for the non-hispanic, black race variable and for marital status seem to have the strongest significance. The sign for the race coefficient is negative, which fits the exposition in Alsan and Wanamaker 2018. Years of education also has some predictive power.

The full set of controls leans on the literature regarding other variables which may predict trust, such as health outcomes, insurace statuses, and pervious marital history. Each of these may have a link with trust in theory, but the coefficient for Depression (and less so for Medicaid coverage) is negative and particularly predictive of trust.

Table 9: Determinants of general trust

	Demographics	Full controls
Female	0.19	0.27
	(0.17)	(0.18)
Years of education	0.05*	0.04
	(0.03)	(0.03)
Married	0.50***	0.17
	(0.18)	(0.19)
NH Black	−0.99***	−1.07***
	(0.22)	(0.23)
Hispanic	−0.38	−0.34
	(0.27)	(0.27)
NH Other	−0.19	−0.19
	(0.32)	(0.33)
Depression		−0.14***
		(0.05)
Health conditions		−0.05
		(0.07)
Covered by Medicare		0.15
		(0.30)
Covered by Medicaid		−0.58**
		(0.28)
Has life insurance		0.25
		(0.18)
Number of reported divorces		−0.16
		(0.11)
Number of reported times being widowed		0.00
		(0.25)
Constant	2.04	2.90
	(1.49)	(1.84)
Observations	894.00	875.00
Adj. R-squared	0.09	0.12
Joint test: Age bins p-value	0.00	0.00
Joint test: Race p-value	0.00	0.00

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Although these are standard control variables, this information is useful as it will help us to understand the distinguish the ability of trust and these different control variables to explain the variability (and potential hump-shape) in income and returns.

4.2 Trust and economic performance

After examining the determinants of trust, I focus the analysis of the empirical relationship between both income and returns and trust on a subset of measures. For income, I omit the analysis for

labor income. I also exclude the analysis of the log of total income. Labor income results are not statistically significant. Given the oversampling of older, retired households, lack of predictive power for trust on labor income is not alarming. Total income results are robust to both log and IHS transformations.

For returns, I omit core and core and retirement portfolio results and instead focus the statistical analysis on returns to net wealth only. The results for the smaller portfolio compositions are largely insignificant across specifications (though the results are more encouraging when retirement assets are excluded). Again, with more retirees in the sample, an analysis of retirement assets may take more creative efforts. That said, I show significant that trust is predictive for the raw measure of returns to net wealth, and the relationship is robust to different trimming and winsorization choices.[8]

4.2.1 Total income and trust

First I test whether the hump-shaped relationship between trust and income in Butler, Giuliano, and Guiso 2016 holds for the 2020 HRS cohort. In Table 4.2.1, we see the coefficients for both the linear and quadratic terms are significant; the predictive power survives even when the additional controls important for explaining income are included. Since the goal is to uncover whether there is a maximal level of trust regarding economic performance, then we care about the significance of both trust coefficients. Here, we reject the hypothesis that both coefficients are 0.[9]

[8] This is important given the difficulty with computing returns in the U.S. due to lack of available data on flows in and out of narrowly defined asset classes.

[9] This rejection of the Wald test for the linear and quadratic trust terms (i.e. reject both coefficients are zero) will be true for most of the models analyzed in this paper.

Table 10: Total income (2020) on general trust

	1	2	3	4
Trust	0.03***	0.19***	0.00	0.05**
	(0.01)	(0.02)	(0.01)	(0.02)
Trust^2		−0.02***		−0.00**
		(0.00)		(0.00)
Female			−0.27***	−0.27***
			(0.04)	(0.04)
Years of education			0.07***	0.07***
			(0.01)	(0.01)
Married			0.14***	0.14***
			(0.04)	(0.04)
Born in U.S.			0.02	0.02
			(0.07)	(0.07)
In labor force			0.50***	0.49***
			(0.05)	(0.05)
NH Black			−0.24***	−0.23***
			(0.05)	(0.05)
Hispanic			−0.24***	−0.23***
			(0.07)	(0.07)
NH Other			−0.27***	−0.27***
			(0.09)	(0.09)
Constant	0.83***	0.53***	−0.48***	−0.51***
	(0.05)	(0.06)	(0.14)	(0.16)
Observations	900.00	900.00	890.00	890.00
Adj. R-squared	0.01	0.05	0.35	0.35
Joint test: Trust p-value	.	0.00	.	0.08
Joint test: Age bins p-value	.	.	0.00	0.00
Joint test: Race p-value	.	.	0.00	0.00

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I also conduct the joint Wald test for the null hypotheses that the coefficients on the age bins and race categories are jointly equal to zero. In both cases, the null hypothesis is rejected at conventional significance levels.

With income data available from 2002-2022, I consider the statistical relationship between average total income and trust. This can be seen in Table 4.2.1. Not only does the pattern persist, but the significance is stronger not only for trust coefficients, but for the other explanatory variables as well. Again, the Wald tests on the coefficients for race and age bins are rejected, meaning each matter for long-run total income.

Table 11: Average total income (avg defl wins, IHS) on General trust (2020)

	1	2	3	4
Trust	0.03***	0.21***	0.00	0.05***
	(0.01)	(0.02)	(0.01)	(0.02)
Trust^2		−0.02***		−0.00***
		(0.00)		(0.00)
Female			−0.25***	−0.25***
			(0.04)	(0.04)
Years of education			0.08***	0.08***
			(0.01)	(0.01)
Married			0.20***	0.19***
			(0.04)	(0.04)
Born in U.S.			−0.03	−0.03
			(0.06)	(0.06)
In labor force			0.37***	0.37***
			(0.04)	(0.04)
NH Black			−0.27***	−0.26***
			(0.05)	(0.05)
Hispanic			−0.29***	−0.28***
			(0.06)	(0.06)
NH Other			−0.28***	−0.29***
			(0.08)	(0.08)
Constant	0.90***	0.58***	−0.62***	−0.66***
	(0.05)	(0.05)	(0.15)	(0.17)
Observations	900.00	900.00	890.00	890.00
Adj. R-squared	0.01	0.07	0.42	0.43
Joint test: Trust+Trust^2 p-value	.	0.00	.	0.03
Joint test: Age bins p-value	.	.	0.00	0.00
Joint test: Race p-value	.	.	0.00	0.00

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It is important to note that this is where the assumption regarding constant trust becomes required. For the previous model, all of the variables are contemporaneous. With average total income as the dependent variable, trust is viewed as a persistent trait about respondents that influences economic decisions over time so that it matters for long-run income. [10]

4.2.2 Returns to net wealth and trust

The presence of hump-shaped empirical relationship between earnings and trust is important because it aligns with previous literature Butler, Giuliano, and Guiso 2016. Given the oversampling of older and retired households by the HRS, this is an important first step towards using the dataset to argue a similar pattern trust and returns.

[10] Lesmeister, Limbach, and Goergen 2019 is an example of work which also assumes trust is time invariant.

For earnings and trust, I used contemporaneous measures to see if the HRS results aligned with the literature. Since returns to net wealth are of primary interest, and the inherent struture of returns and financial contracts is about future payoff/repayment, I use the 2020 trust measure to predict returns in 2022 (the following wave of the survey).

Table 4.2.2 shows the relationship between 2022 returns to net worth and 2020 trust and other control variables. These cross-section results are discouraging in that, the significance of the trust coefficients erodes when control variables are included. The Wald test on the linear and quadratic term for trust also fails to reject the null hypothesis that the coefficients are both zero.

Among the controls which are significant for returns to net wealth, education is positive and has the strongest predictive power. Race and gender matter here, where all racial minority groups and women take a hit to returns for the cross section. I also conduct Wald tests for the null hypothesis that the coefficients on the age bins, wealth deciles, and race categories are jointly equal to zero. In all three cases, the null hypothesis is rejected at conventional significance levels, indicating that these groups of variables contribute significantly to explaining variation in the returns to net wealth.

Table 12: 2022 Returns to net wealth (2022) on General trust (2020) (5 percent winsorized)

	1	2	3	4
Trust	0.01	0.06**	0.01	0.03
	(0.01)	(0.02)	(0.01)	(0.03)
Trust^2		−0.00*		−0.00
		(0.00)		(0.00)
Female			−0.10**	−0.09**
			(0.04)	(0.04)
Years of education			0.03***	0.02***
			(0.01)	(0.01)
Married			0.08	0.08
			(0.05)	(0.05)
Born in U.S.			0.03	0.03
			(0.05)	(0.05)
NH Black			−0.16**	−0.15**
			(0.06)	(0.07)
Hispanic			−0.13*	−0.13*
			(0.07)	(0.07)
NH Other			−0.05	−0.05
			(0.12)	(0.12)
In labor force			0.13**	0.13**
			(0.05)	(0.05)
_cons	0.24***	0.13**	−0.49**	−0.47**
	(0.06)	(0.07)	(0.20)	(0.20)
Observations	497.00	497.00	406.00	406.00
Adj. R-squared	0.00	0.01	0.23	0.23
Joint test: Trust p-value		0.03		0.54
Joint test: Age bins p-value	.	.	0.00	0.00
Joint test: Wealth deciles p-value	.	.	0.00	0.00
Joint test: Race p-value	.	.	0.06	0.07

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The main empirical object of analysis for this paper is the distribution of returns to net worth for respondents in the HRS from 2002 to 2022. The persistent component of returns is of particular interest to the heterogeneous agent literature, and the panel regression version of the model with individual fixed effects will allow us to describe this components. Evidence of individuals persistently earning different returns on average is motivation for allowing households to be heterogeneous in their returns in the macro modeling framework, which I explore in another paper.

I also inspect the relationship between average annual returns over the period and trust in Table 4.2.2. Here the hump-shaped relationship between trust and returns has empirical support. The individual coefficients are significant, and the Wald test for both the linear and quadratic term suggests that the model with both terms included does better at explaining returns than the model without (when both coefficients are zero). Labor force participation is highly significant and positive for average returns.

Among the controls significant for returns to net wealth, education remains a strong predictor, while gender and race also show notable effects in the analysis over average returns for the period. I conduct Wald tests and find that the null hypothesis is rejected for the coefficients of trust and age bins being jointly equal to zero.

Table 13: Average Returns to net wealth (avg) on General trust (2020) (5 percent winsorized)

	1	2	3	4
Trust	0.01**	0.07***	0.01*	0.05***
	(0.00)	(0.01)	(0.01)	(0.02)
Trust^2		−0.01***		−0.00**
		(0.00)		(0.00)
Female			−0.04*	−0.04
			(0.02)	(0.02)
Years of education			0.01	0.01
			(0.00)	(0.00)
Married			−0.01	−0.01
			(0.03)	(0.03)
Born in U.S.			−0.02	−0.02
			(0.04)	(0.04)
NH Black			0.02	0.03
			(0.04)	(0.04)
Hispanic			−0.05	−0.05
			(0.05)	(0.05)
NH Other			0.03	0.03
			(0.07)	(0.07)
In labor force			0.10***	0.10***
			(0.03)	(0.03)
_cons	0.15***	0.04	0.07	0.08
	(0.03)	(0.04)	(0.09)	(0.10)
Observations	750.00	750.00	593.00	593.00
Adj. R-squared	0.01	0.03	0.06	0.07
Joint test: Trust p-value	.	0.00	.	0.01
Joint test: Age bins p-value	.	.	0.05	0.11
Joint test: Wealth deciles p-value	.	.	0.22	0.24
Joint test: Race p-value	.	.	0.53	0.48

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4.3 Returns from 2002-2022

The main empirical analysis of this paper involves computing returns across several years and describing the persistent component of returns. As my motivation is characterizing heterogeneity in the rate of return across individuals, I use model specifcations similar to those in Fagereng et al. 2020. First, a pooled OLS regression with a baseline set of controls. Second, a similar pooled

regression aimed at controlling for risk exposure. Each of these specifications are captupred by the linear panel data regression model

$$r_{i,t}^{net} = \alpha + X_{i,t}'\beta + \varepsilon_{i,t}$$

where $X_{i,t}$ the set of regressors includes both time-varying and time-invariant demographic variables, as well as the trust measure as linear or with both linear and quadratic terms present.

The third specification has two stages. I begin with a panel regression including individual and year fixed effects along with the time-invariant regressors from the second model. This regression model is extended so that the error term $\varepsilon_{i,t}$ can be written as

$$\varepsilon_{i,t} = f_{i,t} + e_{i,t},$$

where $e_{i,t}$ may possibly be serially correlated. Then I regress the estimated individual fixed effect onto the time invariant regressors

$$\hat{f}_{i,t} = \bar{X}_{i,t} + u_{i,t},$$

where $\bar{X}_{i,t}$ is the set of time-invariant regressor (subset of the full set of controls). This will give us a sense of how the persistent component changes with the variables excluded from the first stage.

4.3.1 The pooled regression with constant trust

In Table 4.3.1, I show three versions of the model: no trust term, linear trust term, both linear and quadratic terms. The time-invariant trust measure displays the hump-shaped relationship for returns to net wealth in terms inference on the estimated coefficients and the joint Wald test.

Table 14: Panel: returns to net wealth (5 percent winsorized)

	(1)	(2)	(3)
Years of education	-0.00^{***}	-0.01^{***}	-0.01^{***}
	(0.00)	(0.00)	(0.00)
Female	-0.02^{***}	-0.03^{*}	-0.03^{*}
	(0.00)	(0.02)	(0.02)
NH Black	0.06^{***}	0.05^{*}	0.05^{*}
	(0.00)	(0.03)	(0.03)
Hispanic	0.03^{***}	-0.01	-0.01
	(0.01)	(0.03)	(0.03)
NH Other	0.02^{**}	0.01	0.01
	(0.01)	(0.04)	(0.04)
In labor force	0.03^{***}	0.03^{*}	0.03^{*}
	(0.00)	(0.02)	(0.02)
Married	-0.06^{***}	-0.06^{***}	-0.06^{***}
	(0.00)	(0.02)	(0.02)
Born in U.S.	0.02^{***}	0.03	0.03
	(0.00)	(0.02)	(0.02)
Trust		0.01^{**}	0.03^{***}
		(0.00)	(0.01)
Trust^2			-0.00^{**}
			(0.00)
_cons	0.15	-0.56^{***}	-0.54^{***}
	(0.12)	(0.06)	(0.06)
Observations	102091.00	2899.00	2899.00
Adj. R-squared	0.03	0.04	0.04
Joint test: Trust p-value	.	.	0.00
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.00	0.01	0.01
Joint test: Race p-value	0.00	0.26	0.20
Joint test: Region p-value	0.00	0.09	0.11
Joint test: Year p-value	0.00	0.00	0.00

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Education and marital status also have strong predictive power in the panel setting. Along with the general trust meausre, the Wald tests are rejected for the coefficients on age bins, wealth deciles, and year fixed effects being jointly equal to zero.

4.3.2 Controlling for risk exposure

The key point about the documented persistent component to returns is that, even when controlling for risk preferences and exposure, individuals systematic earn different returns on the same assets[11].

[11] This point is further corroborated by evidence of heterogeneity in returns on safe assets like deposit accounts. An example from the academic literature is Deuflhard, Georgarakos, and Inderst 2014, but one can compare rates

Moreover, the descriptive statistics for mean returns show that returns do not seem stable over the time period (ex. Global Financial Crisis). To account for risk exposure in this environment, I add full interaction terms between shares invested in given asset classes (core, retirement, residential) and year. The regression results for the return measures are given in Table 4.3.2.

offered on deposit account at commercial banks like Bank of America versus local banks and see similar discrepancies in everyday life.

Table 15: Panel Spec 2: returns to net wealth (5 percent winsorized, share x year)

	(1)	(2)	(3)
Years of education	-0.01^{***}	-0.01^{***}	-0.01^{***}
	(0.00)	(0.00)	(0.00)
Female	-0.02^{***}	-0.02	-0.02
	(0.00)	(0.02)	(0.01)
NH Black	0.07^{***}	0.04	0.04
	(0.00)	(0.03)	(0.03)
Hispanic	0.05^{***}	-0.01	-0.01
	(0.01)	(0.03)	(0.03)
NH Other	0.02^{***}	0.03	0.03
	(0.01)	(0.04)	(0.04)
In labor force	0.03^{***}	0.02	0.02
	(0.00)	(0.02)	(0.02)
Married	-0.06^{***}	-0.07^{***}	-0.07^{***}
	(0.00)	(0.02)	(0.02)
Born in U.S.	0.01^{*}	0.02	0.02
	(0.00)	(0.02)	(0.02)
Share core	-0.06^{**}	0.25	0.25
	(0.03)	(0.24)	(0.24)
Share IRA	-0.24^{***}	-0.08	-0.08
	(0.03)	(0.24)	(0.24)
Share residential	-0.13^{***}	-0.00	0.01
	(0.03)	(0.23)	(0.23)
Share long-term debt	-0.02^{***}	-0.15	-0.16
	(0.01)	(0.27)	(0.27)
Share other debt	-0.01	0.13	0.14
	(0.01)	(0.17)	(0.16)
Trust		0.01^{**}	0.03^{***}
		(0.00)	(0.01)
Trust^2			-0.00^{**}
			(0.00)
_cons	0.17	-0.46^{**}	-0.45^{**}
	(0.16)	(0.22)	(0.22)
Observations	100837.00	2866.00	2866.00
Adj. R-squared	0.05	0.06	0.06
Joint test: Trust+Trust^2 p-value	.	.	0.01
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.00	0.03	0.02
Joint test: Race p-value	0.00	0.40	0.31
Joint test: Region p-value	0.00	0.05	0.06
Joint test: Year p-value	0.00	0.00	0.00
Joint test: Share (core, IRA, res) p-value	0.00	0.28	0.29
Joint test: Share (debt long, other) p-value	0.00	0.64	0.58
Joint test: Share core x year p-value	0.00	0.01	0.01
Joint test: Share IRA x year p-value	0.00	0.01	0.01
Joint test: Share res x year p-value	0.00	0.01	0.01
Joint test: Share debt long x year p-value	0.00	0.13	0.13
Joint test: Share debt other x year p-value	0.61	0.00	0.00

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The predictive power of the trust coefficents survives after controlling for risk. Although the uninteracted terms for shares in various asset classes are not signficant, the performance for all asset classes and for the "Other debt" liability regarding the Wald test suggests that portfolio allocation among these assets and debt over the time period is explanatory for the variation in returns.

Education and marital status remain highly predictive as well even after controlling for risk exposure. Along with the general trust measure, the Wald tests are rejected for the coefficients on age bins, wealth deciles, region, and year fixed effects being jointly equal to zero. The interaction terms between asset class shares and year are also jointly significant for all asset classes and for the "Other debt" liabilities class.

4.3.3 Fixed effects and the persistent component of returns

Evidence of a maximal level of trust associated with returns to net wealth is a novel extension of the findings by Butler, Giuliano, and Guiso 2016. I now turn attention to estimating the persistent component of returns by estimated the model with individual fixed effects. Notably, inclusion of this term will require we drop the time-invariant regressors from the set of control variables.

In this setting, the coefficents on the uninteracted share terms for all of the available asset (core, retirement, residential) and liabilities (long, other) are now significant, which can be seen in Table 4.3.3.

Table 16: Panel Spec 3: returns to net wealth (5 percent winsorized, share x year)

	(1)
In labor force	0.03***
	(0.00)
Married	−0.05***
	(0.01)
Share core	0.25***
	(0.04)
Share IRA	0.08**
	(0.04)
Share residential	0.14***
	(0.04)
Share long-term debt	−0.01*
	(0.01)
Share other debt	−0.08**
	(0.04)
_cons	−0.49***
	(0.14)
Observations	101088.00
Within R^2	0.19
ρ	0.66
σ_u	0.46
σ_e	0.33
Joint test: Age bins p-value	0.00
Joint test: Region p-value	0.20
Joint test: Wealth deciles p-value	0.00
Joint test: Year p-value	0.00
Joint test: Share (core, IRA, res) p-value	0.00
Joint test: Share (debt long, other) p-value	0.01

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Employment status and marital status are significant predictors of returns to net wealth when fixed effects are included. Wald tests reject the null hypothesis that the coefficients on age bins, wealth deciles, and year fixed effects are jointly equal to zero, indicating their relevance in explaining within-individual return variation. The asset share variables (core, IRA, residential) and debt shares (long-term and other) are also jointly significant, confirming the importance of portfolio composition even after controlling for individual fixed effects.

In terms of the estimated fixed effect, Table 4.3.3 shows that a large share of total variance is captured by ρ, which is evidence in favor of heterogeneity in returns across respondents. Consistent with this, σ_u (the standard deviation of person-specific intercepts) is sizable, suggesting substantial persistent differences in return levels across individuals.

By contrast, σ_e captures the remaining within-person time variation in returns that is not explained by observed controls. The within-R^2 is therefore interpreted as the share of within-respondent return movements explained by the included time-varying covariates – about only a third

of the within-person variation in returns is left after controlling for observables, and importantly, individual fixed effects.

The distribution of estimated individual fixed effects (demeaned, winsorized top/bottom 1%) from this model is shown in the following figure **??**. The is centered close to zero, which indicates that most households have only modest persistent deviations from the return predicted by observed covariates and year effects. Relative to the Norwegian benchmark, the HRS distribution is visibly more compressed and right-skewed: there is a longer positive tail and fewer large negative fixed effects. This pattern is consistent with a small group of households exhibiting persistently higher net-wealth returns, while most households cluster around near-zero persistent effects.

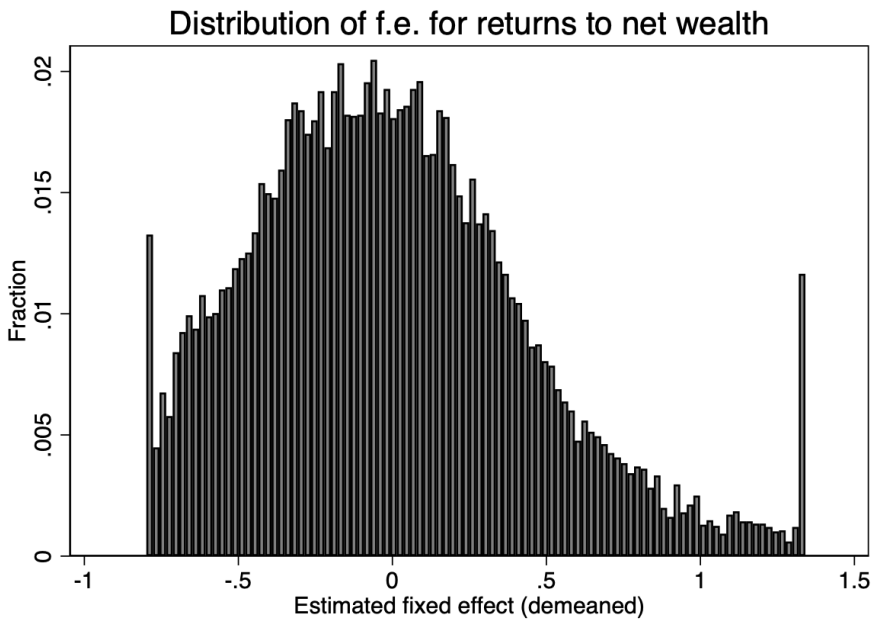

Figure 50: FE distribution: returns to net wealth

In the second stage, I predict the estimated fixed effects using the time-invariantregressors (education, gender, race, born in U.S., trust). The second-stage results for winsorized returns to net wealth are given in Table 4.3.3.

Table 17: Second-stage: FE from returns to net wealth (5 percent winsorized) on time-invariant vars

	(1)	(2)	(3)
Years of education	-0.04^{***}	-0.04^{***}	-0.04^{***}
	(0.00)	(0.01)	(0.01)
Female	0.01^{**}	0.04	0.04
	(0.01)	(0.04)	(0.04)
NH Black	0.30^{***}	0.25^{***}	0.25^{***}
	(0.01)	(0.05)	(0.05)
Hispanic	0.22^{***}	0.23^{***}	0.23^{***}
	(0.01)	(0.05)	(0.05)
NH Other	0.16^{***}	0.08	0.09
	(0.02)	(0.07)	(0.07)
Born in U.S.	0.07^{***}	0.14^{***}	0.14^{***}
	(0.01)	(0.05)	(0.05)
Trust		-0.00	0.02
		(0.01)	(0.02)
Trust^2			-0.00
			(0.00)
_cons	0.39^{***}	0.40^{***}	0.37^{***}
	(0.02)	(0.10)	(0.10)
Observations	21543.00	558.00	558.00
Adj. R-squared	0.14	0.16	0.16
Joint test: Trust p-value	.	.	0.65
Joint test: Race p-value	0.00	0.00	0.00

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The estimated individual fixed effect is the average return for an individual after removing what part of average returns is predicted by the time-varying regressors. Race and education are predictive for the fixed effect, but unfortunately the trust variable is no longer significant for any version of the model.

Although the trust coefficients fail to reject the test that they are equal to 0, I present a figure 51 of the relationship between estimated fixed effects and trust here. Withstanding some observations, it does seem that the maximal levels of the estimated fixed effect are associated with the medium range of trust values between 0 and 10.

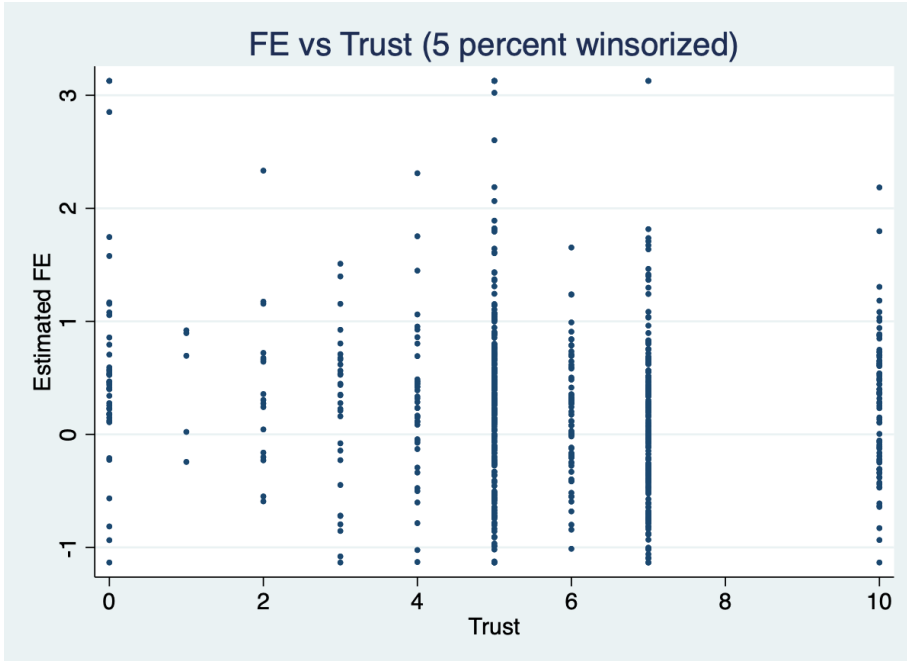

Figure 51: FE vs. trust (5 percent winsorized)

4.4 The "right amount of trust" in the HRS

The model specifications including both a linear and quadratic term for trust are of the form

$$r_{i,t}^{net} = \alpha + \beta_1 \mathit{Trust}_i + \beta_2 \mathit{Trust}_i^2 + X_i'\gamma + \varepsilon_i$$

. Taking the partial derivative with respect to trust is

$$\partial y_i / \partial \mathit{Trust}_i = \beta_1 + 2\beta_2 \mathit{Trust}_i$$

. Setting this to zero yields the critical point $\mathit{Trust}^* = -\beta_1/(2\beta_2)$. When $\hat{\beta}_1 \geq 0, \hat{\beta}_2 \leq 0$ the relationship is hump-shaped (concave) and the estimated level of trust which maximizes returns to net wealth is

$$\mathit{Trust}^* = -\hat{\beta}_1/(2\hat{\beta}_2)$$

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4.4.1 The maximal level of trust for income and returns

Table 4.4.1 reports the turning point Trust^* for each specification, along with $\hat{\beta}_1$, $\hat{\beta}_2$, whether both coefficients are individually significant, and the joint test p-value.

Table 18: Estimated trust *Trust** associated with maximal economic performance

Model	Spec	N	$\hat{\beta}_1$	$\hat{\beta}_2$	Both sig.?	Joint test	(*Trust*)*
Total Income (2022)	Controls	890	0.0494	-0.0044	Yes	0.0766	5.6209
Total Income (Avg.)	Controls	890	0.0494	-0.0046	Yes	0.0278	5.3329
Returns to net wealth (2022)	Controls	406	0.0309	-0.0022	No	0.5431	6.8606
Returns to net wealth (Avg.)	Controls	593	0.0469	-0.0034	Yes	0.0133	6.9121
Panel specification 1	Controls	2,899	0.0279	-0.0019	Yes	0.0018	7.4222
Panel specification 2	Controls	2,866	0.0264	-0.0018	Yes	0.0073	7.3211

Across all six specifications, $\hat{\beta}_1 > 0$ and $\hat{\beta}_2 < 0$, so the relationship is hump-shaped and *Trust** is a maximum. Both trust coefficients are individually significant in five of the six specifications: total income in 2022, average total income, average returns to net worth, and returns to net worth in panel specifications 1 and 2. The Wald test for the linear and quadratic trust terms is rejected at the 5% level in average total income, average returns to net worth, and both panel specifications. Estimated trust *Trust** associated with maximal economic performance ranges from about 5.3 for average total income to 7.4 for returns to net worth in the baseline panel specification, and in every case it lies within the observed trust scale $[0, 10]$.

4.4.2 Delta method for estimated trustassociated with maximal economic performance

Because *Trust** is a nonlinear function of $\hat{\beta}_1$ and $\hat{\beta}_2$, Stata uses the delta method to approximate its standard error. Table 4.4.2 reports *Trust**, SE(*Trust**), 95% confidence intervals, and p-values which test $H_0 : \textit{Trust}^* = 0$.

Table 19: Delta-method results for trust turning points

Model	N	$(Trust)^*$	SE$(Trust)^*$	p$(Trust)^*$	$\underline{CI}$	$\overline{CI}$
Total Income (2022)	890	5.6209	0.8174	0.0000	4.0188	7.2229
Total Income (Avg.)	890	5.3329	0.6858	0.0000	3.9888	6.6770
Returns to net wealth (2022)	406	6.8606	2.8117	0.0147	1.3498	12.3714
Returns to net wealth (Avg.)	593	6.9121	0.9832	0.0000	4.9850	8.8392
Panel spec. 1	2,899	7.4222	1.0247	0.0000	5.4138	9.4306
Panel spec. 2	2,866	7.3211	1.0468	0.0000	5.2693	9.3728

5 Extensions

In this section, I describe the data and accompanying statistical model used to understand the role of *financial literacy* in the current story of an empirical relationship between trust and returns.

Additionally, the literature on trust and social capital is clear on how heterogenity in income, race and other factors related to one's neighborhood can be important for trust. To this end, we consider two additional variables in the survey: respondent *region* and *hometown population size*.

5.1 Financial literacy and returns

Financial literacy can be an important factor when trying to understand the persistent component of returns. The literature on measuring financial literacy using survey data has agreed on three questions (the "Big Three") to measure financial literacy for respondents. The first asks about compound interest: given $100 in a savings account at 2% per year, how much would you have after 5 years—less than $102, exactly $102, or more than $102? The second asks about inflation: if the interest rate is 1% and inflation is 2%, after one year could you buy more, the same, or less than today? The third asks about risk diversification: whether the statement "buying a single company's stock usually provides a safer return than a stock mutual fund" is true or false.

5.1.1 Descriptive statistics

Table 20: Financial literacy summary (2020)

Variable	N	Mean (prop. correct)
Interest	1302	0.74
Inflation	1273	0.76
Risk	1072	0.75

Table 21: Financial literacy: Literacy score (0–3) frequency

Literacy score	N	Pct
0	25	2.4
1	150	14.4
2	364	35.0
3	500	48.1

Table 22: Financial literacy: Inflation by gender

Gender	N	Mean
Male	531	0.79
Female	742	0.75

Table 23: Financial literacy: Inflation by race/ethnicity

Race/ethnicity	N	Mean
White (NH)	729	0.82
Black (NH)	270	0.67
Hispanic	192	0.67
Other (NH)	81	0.77

Table 24: Financial literacy: Inflation by age

Age	N	Mean
25–29	2	0.50
30–34	2	0.00
40–44	7	0.57
45–49	13	0.54
50–54	50	0.74
55–59	249	0.74
60–64	256	0.79
65–69	250	0.78
70–74	154	0.81
75–79	123	0.85
80–84	120	0.68
85–89	46	0.76

Table 25: Financial literacy: Inflation by education

Education	N	Mean
no hs	211	0.67
hs	330	0.74
some college	340	0.75
4yr degree	200	0.83
grad	186	0.87

Table 26: Financial literacy: Literacy score by gender

Gender	N	Mean
Male	471	2.33
Female	568	2.25

Table 27: Financial literacy: Literacy score by race/ethnicity

Race/ethnicity	N	Mean
White (NH)	619	2.45
Black (NH)	209	2.03
Hispanic	138	2.01
Other (NH)	72	2.19

Table 28: Financial literacy: Literacy score by age

Age	N	Mean
25–29	2	1.00
30–34	2	1.50
40–44	6	2.00
45–49	12	1.75
50–54	41	2.49
55–59	207	2.20
60–64	217	2.34
65–69	212	2.33
70–74	130	2.39
75–79	92	2.38
80–84	80	2.08
85–89	37	2.30

Table 29: Financial literacy: Literacy score by education

Education	N	Mean
no hs	147	2.01
hs	267	2.01
some college	269	2.35
4yr degree	177	2.54
grad	173	2.60

Table 5.1.1 shows the cross correlations between the financial literacy measures and the general measure of trust. There is little correlation between trust and any of the financial literacy variables. More worrisome is that there is little correlation amongst the financial literacy variables. If they

are all measuring the same thing, there should be some correlation.

Table 30: Financial literacy and trust correlations

	Interest	Inflation	Risk diversification	General trust
Interest	1.00	-0.01	0.03	0.01
Inflation	-0.01	1.00	0.07	-0.03
Risk diversification	0.03	0.07	1.00	0.04
General trust	0.01	-0.03	0.04	1.00

Interest, inflation, risk diversification, and general trust (2020).

5.1.2 Results

Table 31: Cross section: r5 returns (5 percent wins) on financial literacy and trust

	(1)	(2)	(3)	(4)
Female	−0.11**	−0.12**	−0.11**	−0.12**
	(0.05)	(0.05)	(0.05)	(0.05)
Years education	0.02**	0.02**	0.02**	0.02*
	(0.01)	(0.01)	(0.01)	(0.01)
In labor force	0.12**	0.11**	0.12**	0.11**
	(0.05)	(0.05)	(0.05)	(0.05)
Married	0.06	0.06	0.06	0.06
	(0.05)	(0.05)	(0.06)	(0.06)
Born in U.S.	0.03	0.03	0.02	0.03
	(0.06)	(0.06)	(0.06)	(0.06)
NH Black	−0.19**	−0.19**	−0.18**	−0.18**
	(0.08)	(0.08)	(0.08)	(0.08)
Hispanic	−0.14	−0.14	−0.13	−0.14
	(0.09)	(0.09)	(0.09)	(0.09)
NH Other	0.06	0.07	0.07	0.07
	(0.11)	(0.11)	(0.11)	(0.11)
Interest	0.07		0.07	
	(0.07)		(0.07)	
Inflation	0.15**		0.16**	
	(0.06)		(0.06)	
Risk div	0.01		0.01	
	(0.06)		(0.06)	
Literacy score		0.07*		0.08**
		(0.04)		(0.04)
General trust			0.01	0.01
			(0.03)	(0.04)
(General trust)^2			0.00	−0.00
			(0.00)	(0.00)
_cons	−0.59***	−0.57***	−0.61***	−0.58***
	(0.23)	(0.22)	(0.23)	(0.22)
Observations	319.00	319.00	319.00	319.00
Adj. R-squared	0.25	0.25	0.25	0.25
Joint test: Trust p-value	.	.	0.69	0.80

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Table 32: Average r5 returns (5 percent wins) on financial literacy and trust

	(1)	(2)	(3)	(4)
Female	−0.04	−0.04	−0.05	−0.05*
	(0.03)	(0.03)	(0.03)	(0.03)
Years education	0.00	0.00	0.00	0.00
	(0.01)	(0.01)	(0.01)	(0.01)
In labor force	0.08**	0.08**	0.08**	0.07**
	(0.04)	(0.04)	(0.04)	(0.04)
Married	0.02	0.02	0.02	0.02
	(0.03)	(0.03)	(0.03)	(0.03)
Born in U.S.	−0.03	−0.03	−0.04	−0.04
	(0.04)	(0.04)	(0.04)	(0.04)
NH Black	−0.03	−0.03	−0.01	−0.01
	(0.05)	(0.05)	(0.05)	(0.05)
Hispanic	−0.09	−0.10*	−0.08	−0.09
	(0.06)	(0.06)	(0.06)	(0.06)
NH Other	0.02	0.02	0.03	0.03
	(0.07)	(0.07)	(0.07)	(0.07)
Interest	−0.00		0.00	
	(0.03)		(0.03)	
Inflation	0.08**		0.09**	
	(0.04)		(0.04)	
Risk div	−0.03		−0.03	
	(0.04)		(0.04)	
Literacy score		0.01		0.02
		(0.02)		(0.02)
General trust			0.03	0.03
			(0.02)	(0.02)
(General trust)^2			−0.00	−0.00
			(0.00)	(0.00)
_cons	0.11	0.11	0.09	0.09
	(0.12)	(0.12)	(0.12)	(0.12)
Observations	440.00	440.00	440.00	440.00
Adj. R-squared	0.04	0.03	0.05	0.04
Joint test: Trust p-value	.	.	0.04	0.07

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Table 33: Panel Spec 1 (pooled): r5 returns on financial literacy and trust

	(1)	(2)	(3)	(4)
Years education	−0.00	−0.00	−0.00	−0.00
	(0.00)	(0.00)	(0.00)	(0.00)
Female	−0.02	−0.03	−0.03	−0.03*
	(0.02)	(0.02)	(0.02)	(0.02)
NH Black	0.04	0.04	0.05	0.05
	(0.03)	(0.03)	(0.03)	(0.03)
Hispanic	0.01	0.01	0.02	0.01
	(0.04)	(0.04)	(0.04)	(0.04)
NH Other	0.02	0.02	0.04	0.03
	(0.04)	(0.04)	(0.04)	(0.04)
In labor force	0.04**	0.04*	0.04*	0.03*
	(0.02)	(0.02)	(0.02)	(0.02)
Married	−0.07***	−0.07***	−0.07***	−0.07***
	(0.02)	(0.02)	(0.02)	(0.02)
Born in U.S.	0.01	0.01	0.01	0.01
	(0.03)	(0.03)	(0.03)	(0.03)
Interest	0.02		0.03	
	(0.02)		(0.02)	
Inflation	−0.00		0.00	
	(0.02)		(0.02)	
Risk div	−0.01		−0.01	
	(0.02)		(0.02)	
Literacy score		0.00		0.00
		(0.01)		(0.01)
General trust			0.02	0.01
			(0.01)	(0.01)
(General trust)^2			−0.00	−0.00
			(0.00)	(0.00)
_cons	−0.58***	−0.56***	−0.58***	−0.55***
	(0.06)	(0.06)	(0.06)	(0.06)
Observations	2134.00	2134.00	2134.00	2134.00
Adj. R-squared	0.02	0.02	0.02	0.02
Joint test: Trust p-value	.	.	0.07	0.09
Joint test: Region p-value	0.03	0.03	0.02	0.02

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Table 34: Panel Spec 2 (share x year): r5 returns on financial literacy and trust

	(1)	(2)	(3)	(4)
Years education	−0.01	−0.01	−0.01*	−0.01*
	(0.00)	(0.00)	(0.00)	(0.00)
Female	−0.02	−0.03	−0.03*	−0.03**
	(0.02)	(0.02)	(0.02)	(0.02)
NH Black	0.03	0.02	0.03	0.03
	(0.03)	(0.03)	(0.03)	(0.03)
Hispanic	−0.02	−0.03	−0.02	−0.02
	(0.04)	(0.04)	(0.04)	(0.04)
NH Other	0.04	0.03	0.05	0.05
	(0.05)	(0.05)	(0.04)	(0.05)
In labor force	0.03*	0.03*	0.03*	0.03*
	(0.02)	(0.02)	(0.02)	(0.02)
Married	−0.07***	−0.07***	−0.07***	−0.07***
	(0.02)	(0.02)	(0.02)	(0.02)
Born in U.S.	0.01	0.01	0.01	0.01
	(0.03)	(0.03)	(0.03)	(0.03)
Interest	0.01		0.02	
	(0.02)		(0.02)	
Inflation	0.01		0.01	
	(0.02)		(0.02)	
Risk div	−0.01		−0.02	
	(0.02)		(0.02)	
Literacy score		0.00		0.00
		(0.01)		(0.01)
General trust			0.02	0.01
			(0.01)	(0.01)
(General trust)^2			−0.00	−0.00
			(0.00)	(0.00)
_cons	−0.71***	−0.70***	−0.71***	−0.69***
	(0.21)	(0.22)	(0.21)	(0.21)
Observations	2111.00	2111.00	2111.00	2111.00
Adj. R-squared	0.07	0.07	0.07	0.07
Joint test: Trust p-value	.	.	0.03	0.04
Joint test: Region p-value	0.04	0.03	0.02	0.02

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Table 35: Panel Spec 3 (FE, 2nd stage): r5 returns on financial literacy and trust

	(1)	(2)	(3)	(4)
Years education	0.03***	0.03***	0.03***	0.03***
	(0.00)	(0.00)	(0.00)	(0.00)
Female	−0.07***	−0.07***	−0.07***	−0.07***
	(0.03)	(0.03)	(0.03)	(0.03)
NH Black	−0.19***	−0.19***	−0.19***	−0.19***
	(0.03)	(0.03)	(0.03)	(0.03)
Hispanic	−0.14***	−0.14***	−0.14***	−0.14***
	(0.05)	(0.05)	(0.05)	(0.05)
NH Other	−0.06	−0.06	−0.05	−0.06
	(0.07)	(0.07)	(0.07)	(0.07)
Born in U.S.	−0.09**	−0.09**	−0.09**	−0.09**
	(0.04)	(0.04)	(0.04)	(0.04)
Interest	0.07**		0.07**	
	(0.03)		(0.03)	
Inflation	0.05		0.04	
	(0.04)		(0.03)	
Risk div	0.07**		0.07**	
	(0.03)		(0.03)	
Literacy score		0.06***		0.06***
		(0.02)		(0.02)
General trust			0.04**	0.03**
			(0.02)	(0.02)
(General trust)^2			−0.00**	−0.00**
			(0.00)	(0.00)
_cons	−0.31***	−0.31***	−0.36***	−0.36***
	(0.09)	(0.09)	(0.09)	(0.09)
Observations	417.00	417.00	417.00	417.00
Adj. R-squared	0.27	0.27	0.27	0.27
Joint test: Trust p-value	.	.	0.07	0.10

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5.2 Trust in financial institutions

5.2.1 Composite trust score

Table 36: Financial-institutional trust: summary (2020)

Variable	N	Mean
Banks	990	5.35
Financial advisors	1080	4.29
Mutual funds	1072	4.29
Trust in Fin. Institutions	1234	4.78

Table 37: Cross section: r5 returns (5 percent wins) on financial-institutional trust

	(1)	(2)	(3)
Female	-0.08^{**}	-0.08^{**}	-0.08^{**}
	(0.03)	(0.03)	(0.03)
Years education	0.02^{***}	0.02^{***}	0.02^{***}
	(0.01)	(0.01)	(0.01)
In labor force	0.09^{**}	0.09^{**}	0.09^{**}
	(0.04)	(0.04)	(0.04)
Married	0.06	0.06	0.06
	(0.04)	(0.04)	(0.04)
Born in U.S.	0.01	0.01	0.01
	(0.05)	(0.05)	(0.05)
NH Black	-0.14^{**}	-0.14^{***}	-0.14^{***}
	(0.05)	(0.05)	(0.05)
Hispanic	-0.13^{**}	-0.13^{**}	-0.12^{*}
	(0.06)	(0.06)	(0.06)
NH Other	0.07	0.07	0.07
	(0.10)	(0.10)	(0.10)
Trust		0.00	-0.01
		(0.01)	(0.03)
Trust^2			0.00
			(0.00)
_cons	-0.34^{**}	-0.34^{*}	-0.33^{*}
	(0.17)	(0.17)	(0.18)
Observations	585.00	585.00	585.00
Adj. R-squared	0.25	0.25	0.25
Joint test: Trust p-value	.	.	0.94
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.00	0.00	0.00
Joint test: Race p-value	0.02	0.02	0.02

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Table 38: Average r5 returns (5 percent wins) on financial-institutional trust

	(1)	(2)	(3)
Female	−0.02	−0.02	−0.02
	(0.02)	(0.02)	(0.02)
Years education	0.01**	0.01**	0.01**
	(0.00)	(0.00)	(0.00)
In labor force	0.10***	0.10***	0.10***
	(0.03)	(0.03)	(0.03)
Married	0.02	0.02	0.01
	(0.02)	(0.02)	(0.02)
Born in U.S.	−0.02	−0.02	−0.02
	(0.03)	(0.03)	(0.03)
NH Black	0.01	0.01	0.01
	(0.03)	(0.03)	(0.03)
Hispanic	−0.03	−0.03	−0.03
	(0.04)	(0.04)	(0.04)
NH Other	0.02	0.02	0.02
	(0.06)	(0.06)	(0.06)
Trust		−0.00	0.01
		(0.00)	(0.01)
Trust^2			−0.00
			(0.00)
_cons	0.00	0.01	−0.00
	(0.08)	(0.08)	(0.08)
Observations	833.00	833.00	833.00
Adj. R-squared	0.07	0.07	0.07
Joint test: Trust p-value	.	.	0.20
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.07	0.10	0.10
Joint test: Race p-value	0.82	0.82	0.78

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Table 39: Panel Spec 1 (pooled): r5 returns on financial-institutional trust

	(1)	(2)	(3)
Years education	−0.00	−0.00	−0.00
	(0.00)	(0.00)	(0.00)
Female	−0.01	−0.01	−0.01
	(0.01)	(0.01)	(0.01)
NH Black	0.07***	0.07***	0.07***
	(0.02)	(0.02)	(0.02)
Hispanic	0.06**	0.05**	0.05**
	(0.03)	(0.03)	(0.03)
NH Other	0.03	0.03	0.03
	(0.03)	(0.03)	(0.03)
In labor force	0.03**	0.03**	0.03**
	(0.01)	(0.01)	(0.01)
Married	−0.06***	−0.06***	−0.06***
	(0.01)	(0.01)	(0.01)
Born in U.S.	0.01	0.01	0.01
	(0.02)	(0.02)	(0.02)
Trust		−0.01**	0.00
		(0.00)	(0.01)
Trust^2			−0.00
			(0.00)
_cons	−0.55***	−0.54***	−0.55***
	(0.04)	(0.04)	(0.04)
Observations	4219.00	4219.00	4219.00
Adj. R-squared	0.02	0.02	0.02
Joint test: Trust p-value	.	.	0.06
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.00	0.00	0.00
Joint test: Race p-value	0.00	0.01	0.01
Joint test: Region p-value	0.07	0.10	0.10

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Table 40: Panel Spec 2 (share x year): r5 returns on financial-institutional trust

	(1)	(2)	(3)
Years education	-0.01^{**}	-0.01^{**}	-0.01^{**}
	(0.00)	(0.00)	(0.00)
Female	-0.01	-0.01	-0.01
	(0.01)	(0.01)	(0.01)
NH Black	0.04^{**}	0.04^{*}	0.04^{*}
	(0.02)	(0.02)	(0.02)
Hispanic	0.02	0.02	0.02
	(0.03)	(0.03)	(0.03)
NH Other	0.03	0.03	0.03
	(0.03)	(0.03)	(0.03)
In labor force	0.03^{**}	0.03^{**}	0.03^{**}
	(0.01)	(0.01)	(0.01)
Married	-0.05^{***}	-0.05^{***}	-0.05^{***}
	(0.01)	(0.01)	(0.01)
Born in U.S.	0.01	0.01	0.01
	(0.02)	(0.02)	(0.02)
Trust		-0.01^{**}	-0.00
		(0.00)	(0.01)
Trust^2			-0.00
			(0.00)
_cons	-0.56^{***}	-0.55^{***}	-0.56^{***}
	(0.20)	(0.20)	(0.20)
Observations	4219.00	4219.00	4219.00
Adj. R-squared	0.07	0.07	0.07
Joint test: Trust p-value	.	.	0.06
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.00	0.00	0.00
Joint test: Race p-value	0.19	0.25	0.26
Joint test: Region p-value	0.03	0.04	0.04
Joint test: Year p-value	0.00	0.00	0.00

.

Table 41: Panel Spec 3 (FE, 2nd stage): r5 returns on financial-institutional trust

	(1)	(2)	(3)
Years education	0.03***	0.03***	0.03***
	(0.00)	(0.00)	(0.00)
Female	−0.05**	−0.05***	−0.05**
	(0.02)	(0.02)	(0.02)
NH Black	−0.22***	−0.20***	−0.19***
	(0.03)	(0.03)	(0.03)
Hispanic	−0.17***	−0.15***	−0.16***
	(0.03)	(0.03)	(0.03)
NH Other	−0.06	−0.04	−0.04
	(0.06)	(0.06)	(0.06)
Born in U.S.	−0.06	−0.06	−0.06
	(0.04)	(0.04)	(0.04)
Trust		0.02***	0.06***
		(0.00)	(0.02)
Trust^2			−0.00***
			(0.00)
_cons	−0.21***	−0.31***	−0.36***
	(0.06)	(0.06)	(0.07)
Observations	795.00	795.00	795.00
Adj. R-squared	0.23	0.26	0.26
Joint test: Trust p-value	.	.	0.00
Joint test: Race p-value	0.00	0.00	0.00

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5.2.2 Principal component analysis

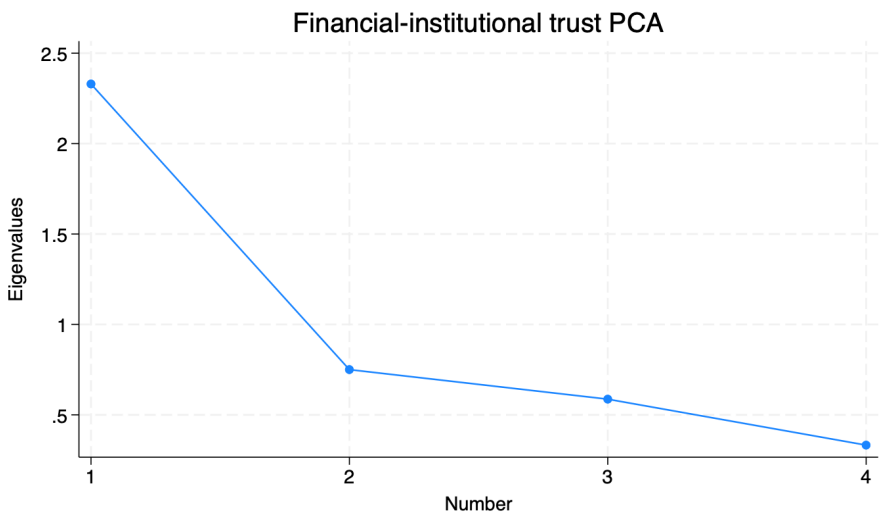

Figure 52: Financial-institutional trust PCA: scree plot

Table 42: Financial-institutional trust PCA: loadings by variable (4 vars: general, banks, advisors, mutual funds)

Variable	PC1	PC2
General trust	0.407	0.879
Banks	0.501	0.012
Financial advisors	0.557	-0.334
Mutual funds	0.523	-0.340

Table 43: Cross section: r5 returns (5 percent wins) on financial-institutional trust

	(1)	(2)	(3)
Female	-0.12^{**}	-0.13^{**}	-0.13^{**}
	(0.05)	(0.05)	(0.05)
Years education	0.02^{**}	0.02^{**}	0.02^{**}
	(0.01)	(0.01)	(0.01)
In labor force	0.11^{*}	0.11^{*}	0.11^{*}
	(0.06)	(0.06)	(0.06)
Married	0.10	0.10	0.10
	(0.06)	(0.06)	(0.06)
Born in U.S.	0.06	0.07	0.07
	(0.07)	(0.07)	(0.07)
NH Black	-0.14^{*}	-0.14^{*}	-0.14^{*}
	(0.08)	(0.08)	(0.08)
Hispanic	-0.13	-0.13	-0.13
	(0.11)	(0.11)	(0.11)
NH Other	-0.01	0.00	0.00
	(0.13)	(0.13)	(0.13)
Trust		0.01	0.01
		(0.02)	(0.02)
Trust^2			0.00
			(0.01)
_cons	-0.40^{*}	-0.36	-0.38
	(0.24)	(0.24)	(0.23)
Observations	271.00	271.00	271.00
Adj. R-squared	0.28	0.28	0.28
Joint test: Trust p-value	.	.	0.70
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.00	0.00	0.00
Joint test: Race p-value	0.23	0.27	0.27

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Table 44: Average r5 returns (5 percent wins) on financial-institutional trust

	(1)	(2)	(3)
Female	−0.05	−0.05	−0.05
	(0.03)	(0.03)	(0.03)
Years education	0.01	0.01	0.01
	(0.01)	(0.01)	(0.01)
In labor force	0.11***	0.11***	0.10**
	(0.04)	(0.04)	(0.04)
Married	−0.02	−0.02	−0.03
	(0.04)	(0.04)	(0.04)
Born in U.S.	0.01	0.01	0.01
	(0.05)	(0.05)	(0.05)
NH Black	0.00	0.00	0.01
	(0.05)	(0.05)	(0.05)
Hispanic	−0.02	−0.02	−0.02
	(0.06)	(0.06)	(0.06)
NH Other	0.02	0.02	0.02
	(0.09)	(0.09)	(0.09)
Trust		−0.00	0.00
		(0.01)	(0.01)
Trust^2			−0.01**
			(0.01)
_cons	0.04	0.04	0.12
	(0.12)	(0.12)	(0.12)
Observations	411.00	411.00	411.00
Adj. R-squared	0.05	0.05	0.06
Joint test: Trust p-value	.	.	0.09
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.02	0.02	0.01
Joint test: Race p-value	0.99	0.99	0.97

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Table 45: Panel Spec 1 (pooled): r5 returns on financial-institutional trust

	(1)	(2)	(3)
Years education	−0.00	−0.00	−0.00
	(0.00)	(0.00)	(0.00)
Female	−0.02	−0.02	−0.02
	(0.02)	(0.02)	(0.02)
NH Black	0.04	0.04	0.04
	(0.03)	(0.03)	(0.03)
Hispanic	0.03	0.03	0.03
	(0.04)	(0.04)	(0.04)
NH Other	−0.03	−0.03	−0.03
	(0.04)	(0.04)	(0.04)
In labor force	0.03	0.03	0.03
	(0.02)	(0.02)	(0.02)
Married	−0.07***	−0.07***	−0.07***
	(0.02)	(0.02)	(0.02)
Born in U.S.	0.03	0.03	0.03
	(0.03)	(0.03)	(0.03)
Trust		−0.01	−0.00
		(0.01)	(0.01)
Trust^2			−0.00
			(0.00)
_cons	−0.58***	−0.59***	−0.56***
	(0.06)	(0.06)	(0.07)
Observations	1864.00	1864.00	1864.00
Adj. R-squared	0.02	0.02	0.02
Joint test: Trust p-value	.	.	0.19
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.08	0.08	0.08
Joint test: Race p-value	0.39	0.41	0.39
Joint test: Region p-value	0.04	0.05	0.07

.

Table 46: Panel Spec 2 (share x year): r5 returns on financial-institutional trust

	(1)	(2)	(3)
Years education	-0.01^{*}	-0.01	-0.01^{*}
	(0.00)	(0.00)	(0.00)
Female	-0.02	-0.02	-0.02
	(0.02)	(0.02)	(0.02)
NH Black	0.02	0.02	0.02
	(0.03)	(0.03)	(0.03)
Hispanic	-0.01	-0.01	-0.01
	(0.04)	(0.04)	(0.04)
NH Other	-0.01	-0.01	-0.01
	(0.04)	(0.04)	(0.04)
In labor force	0.03	0.03	0.03
	(0.02)	(0.02)	(0.02)
Married	-0.06^{**}	-0.06^{**}	-0.06^{**}
	(0.02)	(0.02)	(0.02)
Born in U.S.	0.03	0.03	0.03
	(0.03)	(0.03)	(0.03)
Trust		-0.01	-0.00
		(0.01)	(0.01)
Trust^2			-0.00
			(0.00)
_cons	-0.48^{*}	-0.50^{*}	-0.47^{*}
	(0.27)	(0.27)	(0.27)
Observations	1864.00	1864.00	1864.00
Adj. R-squared	0.06	0.06	0.06
Joint test: Trust p-value	.	.	0.23
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.21	0.22	0.21
Joint test: Race p-value	0.80	0.81	0.79
Joint test: Region p-value	0.04	0.04	0.06
Joint test: Year p-value	0.02	0.02	0.02

.

Table 47: Panel Spec 3 (FE, 2nd stage): r5 returns on financial-institutional trust

	(1)	(2)	(3)
Years education	0.02***	0.02***	0.02***
	(0.00)	(0.00)	(0.00)
Female	−0.04	−0.06**	−0.06*
	(0.03)	(0.03)	(0.03)
NH Black	−0.24***	−0.22***	−0.21***
	(0.04)	(0.04)	(0.04)
Hispanic	−0.19***	−0.18***	−0.19***
	(0.05)	(0.05)	(0.05)
NH Other	−0.07	−0.05	−0.05
	(0.07)	(0.07)	(0.08)
Born in U.S.	−0.07	−0.07	−0.08
	(0.06)	(0.06)	(0.06)
Trust		0.04***	0.05***
		(0.01)	(0.01)
Trust^2			−0.01**
			(0.00)
_cons	−0.03	−0.01	0.03
	(0.07)	(0.07)	(0.07)
Observations	384.00	384.00	384.00
Adj. R-squared	0.17	0.20	0.21
Joint test: Trust p-value	.	.	0.00
Joint test: Race p-value	0.00	0.00	0.00

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Table 48: Cross section: r5 returns (5 percent wins) on financial-institutional trust

	(1)	(2)	(3)
Female	-0.12^{**}	-0.12^{**}	-0.12^{**}
	(0.05)	(0.05)	(0.05)
Years education	0.02^{**}	0.02^{**}	0.02^{**}
	(0.01)	(0.01)	(0.01)
In labor force	0.11^{*}	0.11^{*}	0.11^{*}
	(0.06)	(0.06)	(0.06)
Married	0.10	0.10	0.10
	(0.06)	(0.06)	(0.06)
Born in U.S.	0.06	0.06	0.06
	(0.07)	(0.07)	(0.07)
NH Black	-0.14^{*}	-0.14^{*}	-0.14^{*}
	(0.08)	(0.08)	(0.08)
Hispanic	-0.13	-0.13	-0.14
	(0.11)	(0.11)	(0.11)
NH Other	-0.01	-0.01	-0.01
	(0.13)	(0.13)	(0.13)
Trust		0.01	0.01
		(0.03)	(0.03)
Trust^2			0.01
			(0.02)
_cons	-0.40^{*}	-0.39	-0.41^{*}
	(0.24)	(0.24)	(0.25)
Observations	271.00	271.00	271.00
Adj. R-squared	0.28	0.28	0.28
Joint test: Trust p-value	.	.	0.84
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.00	0.00	0.00
Joint test: Race p-value	0.23	0.25	0.25

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Table 49: Average r5 returns (5 percent wins) on financial-institutional trust

	(1)	(2)	(3)
Female	−0.05	−0.05	−0.05
	(0.03)	(0.03)	(0.03)
Years education	0.01	0.01	0.01
	(0.01)	(0.01)	(0.01)
In labor force	0.11***	0.11***	0.11***
	(0.04)	(0.04)	(0.04)
Married	−0.02	−0.02	−0.02
	(0.04)	(0.04)	(0.04)
Born in U.S.	0.01	−0.00	−0.00
	(0.05)	(0.05)	(0.05)
NH Black	0.00	0.01	0.01
	(0.05)	(0.05)	(0.05)
Hispanic	−0.02	−0.01	−0.01
	(0.06)	(0.06)	(0.06)
NH Other	0.02	0.02	0.02
	(0.09)	(0.09)	(0.09)
Trust		0.05***	0.05***
		(0.02)	(0.02)
Trust^2			0.00
			(0.01)
_cons	0.04	0.10	0.09
	(0.12)	(0.12)	(0.12)
Observations	411.00	411.00	411.00
Adj. R-squared	0.05	0.07	0.07
Joint test: Trust p-value	.	.	0.02
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.02	0.03	0.03
Joint test: Race p-value	0.99	0.97	0.97

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Table 50: Panel Spec 1 (pooled): r5 returns on financial-institutional trust

	(1)	(2)	(3)
Years education	−0.00	−0.00	−0.00
	(0.00)	(0.00)	(0.00)
Female	−0.02	−0.02	−0.02
	(0.02)	(0.02)	(0.02)
NH Black	0.04	0.05*	0.05*
	(0.03)	(0.03)	(0.03)
Hispanic	0.03	0.03	0.03
	(0.04)	(0.04)	(0.04)
NH Other	−0.03	−0.02	−0.02
	(0.04)	(0.04)	(0.04)
In labor force	0.03	0.03	0.03
	(0.02)	(0.02)	(0.02)
Married	−0.07***	−0.07***	−0.07***
	(0.02)	(0.02)	(0.02)
Born in U.S.	0.03	0.02	0.02
	(0.03)	(0.03)	(0.03)
Trust		0.03***	0.03***
		(0.01)	(0.01)
Trust^2			−0.00
			(0.01)
_cons	−0.58***	−0.54***	−0.54***
	(0.06)	(0.06)	(0.06)
Observations	1864.00	1864.00	1864.00
Adj. R-squared	0.02	0.02	0.02
Joint test: Trust p-value	.	.	0.02
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.08	0.06	0.06
Joint test: Race p-value	0.39	0.27	0.27
Joint test: Region p-value	0.04	0.03	0.03

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Table 51: Panel Spec 2 (share x year): r5 returns on financial-institutional trust

	(1)	(2)	(3)
Years education	-0.01^{*}	-0.01	-0.01
	(0.00)	(0.00)	(0.00)
Female	-0.02	-0.02	-0.02
	(0.02)	(0.02)	(0.02)
NH Black	0.02	0.03	0.03
	(0.03)	(0.03)	(0.03)
Hispanic	-0.01	-0.01	-0.01
	(0.04)	(0.04)	(0.04)
NH Other	-0.01	-0.00	-0.01
	(0.04)	(0.04)	(0.04)
In labor force	0.03	0.03	0.03
	(0.02)	(0.02)	(0.02)
Married	-0.06^{**}	-0.06^{**}	-0.06^{**}
	(0.02)	(0.02)	(0.02)
Born in U.S.	0.03	0.02	0.02
	(0.03)	(0.03)	(0.03)
Trust		0.02^{**}	0.02^{**}
		(0.01)	(0.01)
Trust^2			0.00
			(0.01)
_cons	-0.48^{*}	-0.45^{*}	-0.45^{*}
	(0.27)	(0.27)	(0.27)
Observations	1864.00	1864.00	1864.00
Adj. R-squared	0.06	0.06	0.06
Joint test: Trust p-value	.	.	0.06
Joint test: Age bins p-value	0.00	0.00	0.00
Joint test: Wealth deciles p-value	0.21	0.15	0.15
Joint test: Race p-value	0.80	0.69	0.69
Joint test: Region p-value	0.04	0.03	0.03
Joint test: Year p-value	0.02	0.01	0.01

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Table 52: Panel Spec 3 (FE, 2nd stage): r5 returns on financial-institutional trust

	(1)	(2)	(3)
Years education	0.02***	0.02***	0.02***
	(0.00)	(0.00)	(0.00)
Female	−0.04	−0.05	−0.05
	(0.03)	(0.03)	(0.03)
NH Black	−0.24***	−0.26***	−0.25***
	(0.04)	(0.04)	(0.04)
Hispanic	−0.19***	−0.19***	−0.19***
	(0.05)	(0.05)	(0.05)
NH Other	−0.07	−0.08	−0.07
	(0.07)	(0.07)	(0.07)
Born in U.S.	−0.07	−0.05	−0.04
	(0.06)	(0.06)	(0.06)
Trust		−0.07***	−0.06***
		(0.02)	(0.02)
Trust^2			−0.02*
			(0.01)
_cons	−0.03	−0.04	−0.00
	(0.07)	(0.07)	(0.07)
Observations	384.00	384.00	384.00
Adj. R-squared	0.17	0.20	0.20
Joint test: Trust p-value	.	.	0.00
Joint test: Race p-value	0.00	0.00	0.00

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